

Taxing What the State Could See: The Design of Soviet Grain Procurement and the Famine of 1933*

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Abstract

A state that cannot verify its producers' harvest taxes what it can see. Soviet grain procurement was written on sown area, past deliveries, and a yield assessed from the standing crop; from 1933 the machine-tractor station took a second claim, in kind, from the same harvest. Was that design sufficient to leave collective farms below subsistence and to account for the mortality of 1933? In the state's own post-harvest accounts, nine of ten Ukrainian collective farms were left with less than subsistence in 1932, and the food left where policy set it predicts the 1933 mortality rise under every within-region allocation consistent with the printed distributions. The state taxed the base it could observe and accepted the loss; with a subsistence floor beneath the producer, that loss fell on survival. The schedules make no reference to nationality, and the regional gradient of extraction disappears once their terms are accounted for.

*All estimates are computed from the dynamic (*dinamika*) surveys of collective farms run by TsUNKhU of Gosplan, from the surviving primary forms of the same survey for 272 collective farms in Sverdlovsk oblast, from the district handbooks for Ukraine, the Urals and the Lower Volga, and from the replication archive of Markevich et al. (2025). Sources are listed in Tables 3 and 4. All errors are mine.

1 Introduction

A tax on output requires the taxing authority to know the output. Where producers can conceal the crop and the state cannot audit them, the feasible tax is written on what an inspector can verify — the land sown, the deliveries of past years, a yield read off the standing crop — and its burden falls not in proportion to what a producer has but in proportion to what the state can see. The economics of the substitution are known: a claim fixed on a verifiable base transfers the risk of the harvest to the producer (Weitzman, 1974), and a tax administration that cannot observe its intended base shifts to the proxies it can enforce (Best et al., 2015). Those literatures price the substitution in forgone output. Where the taxed producer lives on the residual of his own harvest, its price is survival.

Soviet grain procurement in the early 1930s was a tax of this kind. Until 1933 a collective farm nominally owed a share of its crop, but the size of the obligation followed the procurement task apportioned downward from the centre, not the harvest (Il'inykh, 2015); the law of January 1933 made the base explicit — a fixed quantity per sown hectare, at a norm published for each district (the *raion*, the administrative tier below the region) before the spring sowing. From February 1933 the fee of the machine-tractor station — the state tractor enterprise that worked collective farms' fields for payment — became a second claim in grain on the same harvest, where it had been a money price. Did this design — the verifiable base, the carry-forward of the claim, its rigidity to the realized harvest by rule, the second claim beside the first — account for collective farms being left below subsistence in 1932 and for the mortality of 1933? The materials for an answer survive: the published statutes, the assigned district plans, and the internal statistical record of how each harvest was divided.

According to the statutes, two claims lay on a collective farm's harvest. The obligatory delivery was, until January 1933, a share of the crop, revisable while the campaign ran; the law of 19 January 1933 replaced it with the per-hectare norm, set for each district by the regional executive committee and declared not subject to change. The station's fee was, until 1933, a money price at fixed rates for each job; the decree of 5 February 1933 converted it into a share of a harvest that a commission — with the station's own director among its members — assessed from the standing crop before the harvest was gathered. By 1934 both claims were quantities per hectare, owed to two agencies with separate accounts and no rule for netting one against the other. Relief in crop failure existed on the delivery side, as rationed funds of discounts voted at the centre; the fee statutes contained no provision for it.

The data against which these rules can be checked are the state's internal statistical record. The dynamic surveys of collective farms — restricted typescripts compiled after each harvest from the farms' annual reports — print the division of the crop, delivery, seed, station fee and distribution

to members, for twenty regions, separately for served and unserved farms; the surviving primary forms of the same survey cover 272 farms of Sverdlovsk oblast, for the harvests of 1933 and 1934; district handbooks for Ukraine, the Urals and the Lower Volga print plans and sown areas; district mortality comes from the replication archive of Markevich et al. (2025). These sources record the barn crop, taken into store after the harvest, while the claims were computed before it, so what a farm was left with can be measured against what it owed. They do not record everything: the harvest and its division are observed by region, and at farm level in one oblast only; ethnicity and mortality are observed by district; no source prints a district harvest. At what level, then, is the link from policy to mortality identified, and what can be said beneath that level? The answer requires following the claim from its assignment onward.

The procurement task was set at the centre and divided downward, onto regions, districts, village soviets and farms (Il'inykh, 2015). What determined a district's figure? In 1930 the centre divided the total among regions by the growth of their sown area (Kondrashin, 2015) — a capacity criterion; the planning literature predicts a different rule, the ratchet, in which a fulfilled target is raised and the target follows performance (Berliner, 1957; Weitzman, 1980; Freixas et al., 1985). The district plans of three Ukrainian campaigns survive beside the procurement each district delivered, so the two rules can be run against each other. Each campaign's plan follows the district's most recent realized procurement almost one for one; the district's land, households and draft power explain the earliest plan on record and are displaced by the district's own delivery record thereafter (Section 4). A district's obligation was its own past, carried forward — and a number carried forward from the past cannot respond to the harvest in front of it.

A claim assigned this way met the harvest of 1932, which failed. Under the proportional rule the claim should have fallen with the crop; written as a quantity, it could not. In the regions on the producing list, the slope of the delivery on the yield falls from the statutory share in 1931 to nearly flat by 1933, and across the 1932–33 transition the claim per hectare did not follow the yield's change — the movement was absorbed by the share of the harvest surrendered (Section 4). Relief existed, but as administration, not as rule: reductions were granted from rationed central funds, targeted — by their own text — at the districts whose harvests had suffered most, and where the cut was deepest the reduced quota was then collected in full; after mid-1933 a reduction was a deferral, recoverable from the next harvest. No yield, however low, changed a norm by itself.

The reform of 1933 also changed what the station took. Until then a farm's payment to the station was a cost in money; from February 1933 it was grain from the same harvest on which the delivery stood — and the delivery law itself set separate, lower norms for served farms, so the grain fee and the delivery relief arrived together, and whether a served farm's total burden rose or fell depends on which was larger. Measured together in the surveys' eighty region-by-service cells, before and after the reform, the fee exceeded the delivery relief: the total take from a served farm

rose by roughly four and a half points of the gross harvest (Section 5). The farm accounts, which record the mechanism in the harvests of 1933 and 1934, show the two claims written on different bases — the fee on the farm’s own realized crop, the delivery on the district norm — and a served farm surrendering more per hectare than its unserved neighbour, by roughly the margin the fee’s designer had demanded in advance.

Claims of this size raise the question the famine poses: were farms left below subsistence because their fields failed, or because of how the harvest was divided? The surveys allow the same farms to be counted twice on one accounting chain — once on what they grew, once on what remained after the claims and the farm’s own seed and fodder reserves. In the median region almost no farm grew too little to feed its members; after the division, most fell below the line. Regressing the rise in regional mortality on each count, the harvest-failure count predicts little, the count after the division carries the gradient, and the two are uncorrelated across regions (Section 6). For the 1932 harvest, the state’s claims, not the crops, account for nearly the whole of the rise in Ukrainian collective farms below subsistence.

The gradient of mortality on the food left, estimated between regions, invites the objection that the allocation among a region’s districts, which no source records, could produce it. Section 6 closes by bounding that allocation instead of assuming it, in the partial-identification tradition of Cross and Manski (2002): over every arrangement of the residual consistent with the printed distributions, the food left where policy set it predicts the district mortality rise; holding every region at the least-deprived observed residual prices the gradient at between roughly a quarter and three fifths of the excess deaths recorded across the compared districts; with region effects the association vanishes, and no admissible arrangement explains an appreciable share of mortality within regions. A recalibration to the recovery harvest, when the timing of the campaign requires the gradient to disappear, finds none. The answer to the level question follows: the link is identified between regions, where the lists, the rates and the carry-forward varied; beneath the region the allocation is unidentified, and nothing measurable is lost by estimating where the policy varied.

What the design does not account for is the higher mortality of the Ukrainian districts. Section 7 establishes where that differential did not travel. The Ukrainian premium in delivery disappears once producing status is accounted for; the schedules carry no ethnic term; and the mortality gradient in the Ukrainian population share survives oblast effects and is invariant to every admissible food allocation. Within Ukraine, the association between collectivization and mortality reverses sign from oblast to oblast and is moderated by a district’s capacity, not by its Ukrainian share. The differential is therefore a bounded residual that did not run through the grain quota; its candidate instruments — the conditioning of relief, the restriction of movement, the local campaign — are named in the documents and measured in none of these sources.

The argument stands in the line of Meng et al. (2015), in which an inflexible procurement

system, set on past output, converted a production shock into the famine mortality of rural China, identified between regions because that is the level at which the policy varied. Their inflexibility is inferred from aggregate production and mortality under a model of the planner; in the Soviet records it is observed — in the statutes, in the assignment rule estimated from the plans themselves, in the assessed yield set against the barn crop. Markevich et al. (2025) attribute roughly half of the Ukrainian mortality differential to bias in centrally planned procurement and the remainder to instruments outside the quota — the restriction of movement, the blacklisting of farms and districts, the end of Ukrainization. The bounds revise the division of that attribution, not its total: within regions the quota’s records leave the differential intact under every admissible food allocation, so the procurement half is relocated to the instruments their account already names. The factual frame throughout rests on the historiography that established the famine’s policy character (Davies, 1980; Davies and Wheatcroft, 2004) and on the source studies of the yield assessment and the farm surveys (Zelenin, 1992).

The claim is one of sufficiency, not exclusivity. The design accounts for collective farms left below subsistence with no discretionary cruelty required at any node; it does not account for the Ukrainian mortality differential, which survives every instrument these sources measure. The two claims and their statutory bases are set out in Section 2 and the data in Section 3; Section 4 follows the quota from its assignment to the division of the harvest, Section 5 measures the station’s fee and the total take, Section 6 carries what farms were left to the mortality of 1933 and bounds the allocation beneath the region, Section 7 measures the residual within regions, and Section 8 concludes.

2 Institutional Background

The obligatory delivery. A Soviet collective farm—a *kolkhoz*, a producer cooperative whose members worked land and equipment held in common and were paid out of whatever the farm had left once its obligations were met—owed grain to the state before it paid anyone. Through 1932 that obligation was written as a share of the harvest: grain-producing regions owed between one-quarter and one-third of output, consuming regions no more than one-eighth, and the remainder was nominally “at the full disposal of the *kolkhoz*” (Sovnarkom SSSR, 1930). The share was a ceiling rather than the operative burden. Quotas were in fact set by apportionment from above (Section 4), with separate sub-quotas for state farms, collective farms and individual peasants (Davies, 1980, ch. 9).

The law of 19 January 1933 abolished that system for grain and put the delivery on a new base. A farm owed a fixed quantity for each hectare of its planned spring sowing and each hectare of its actual winter sowing, with the regional averages set separately for farms served and not served by a

machine-tractor station (Sovnarkom SSSR and TsK VKP(b), 1933b). A farm that sowed less than its plan therefore owed no less. The rates followed eight days later, stated in centners per hectare and varying by crop, by region, and by whether the farm was served by a station (Sovnarkom SSSR, 1933b). The regional dimension of the schedule was a region's classification as grain-producing or grain-consuming. For rye on an unserved farm the rate was 3.8 centners per hectare in the Central Black Earth region, 3.1 in Ukraine and 3.0 in West Siberia, against 1.8 in Moscow region and 0.6 in Leningrad region.

The regional average was not what a farm faced. Under article 4 of the January law the executive committee of each region fixed a norm for every district in it, free to go above or below the regional average so long as the regional total held. The district norm was binding on every collective farm of that district and was not subject to change, and the district executive committee computed each farm's obligation from it. The level at which the delivery was permitted to discriminate between farms was therefore the district, by statute.

Two features of the reformed rule bear on everything that follows. Because the obligation was a quantity per hectare and not a fraction of output, a shortfall in the harvest left it unchanged, so the whole of the shortfall fell on what the farm retained. And because the schedule set a lower rate for farms served by a station, the delivery arm was written to fall for exactly the farms that the second arm was about to reach.

The machine-tractor station. A machine-tractor station (MTS) was a state enterprise, legally separate from the farms, that held tractors, threshers and combine harvesters and hired them out to the collective farms of its district. It kept its own accounts and answered to its own plan. Through 1932 a farm paid its station in cash at fixed rates per unit of work performed—so much per hectare plowed, per hectare mown, per unit threshed. In the 1932 grain accounts the line for payment in kind is empty, because there was nothing to enter on it: the station was paid in money. That makes 1932 a clean year to measure against.

From paying for work to paying out of the crop. A year later a farm no longer paid the station for what the station had done. It paid a share of what the farm had grown. That is the change this paper turns on, because it moves the claim off a quantity the farm and the station could both count—hectares plowed, tons threshed—and onto a quantity neither of them could settle between themselves, the size of the harvest. Once the fee is a percentage, the fee depends on somebody's estimate of the crop, and the paragraphs below are about whose estimate that was.

Decree No. 346 of 5 February 1933 made the change in a single article, and named its two halves separately (Sovnarkom SSSR, 1933a):

“In abolition of the existing system of payment by collective farms for the work of the machine-

tractor station *at fixed rates*, payment for the work of the machine-tractor station is established *as determined percentages of the collective farm's harvest*. In abolition of the existing system of payment by collective farms for the work of the machine-tractor station *in money*, payment is established *in kind*.” (art. 9)

The base moved from the work the station had done to the crop the farm had grown, and the payment moved from money to grain. Article 11 set the percentages operation by operation: spring plowing and the raising of fallow at 10 per cent of the harvest, autumn plowing at 9, sowing at 3, mowing at 4 and threshing at 8. Article 10's model contract priced the whole package at 20 per cent. A farm that bought operations separately therefore paid up to 25 per cent where a farm that took everything paid 20.

Article 12 named the body that determined the harvest those percentages were taken of:

“The gross grain harvest of each collective farm ... on the basis of the yield established by the inter-district state commission *for the given district*, is determined by a special commission composed of: the chairman of the district executive committee, the district plenipotentiary of the Committee for Procurement, *the director of the corresponding machine-tractor station*, and the chairman of the collective farm.” (art. 12)

A farm's gross harvest was thus not read off its own books. It was computed from a yield assessed for a whole district by an inter-district state commission and applied to the farm's own sown area, by a local commission of four in which the director of the agency being paid sat on the body that fixed the base of its own fee.

Article 8 required the station to buy up the collective farms' own complex threshers and locomobiles. A farm that no longer owned the machinery to thresh its own grain could not decline the threshing charge.

The state could not see the harvest, so it measured the field instead. A grain quota is easy to write and hard to collect, because the quantity it is written against sits in somebody else's barn. The Soviet state took the view that farms concealed what they had, and it built its accounting system around that belief rather than around an audit. From 1933 the gross harvest of the whole country was determined not from farm records but by a new body, the Central State Yield Commission, working through inter-district commissions in the field. Its instruction of 22 May 1933 told them to measure the crop *on the root*, at the ripening stage, before the harvest: to lay a metre frame on the field at regular intervals, cut the ears alone with a sickle or shears, dry them and thresh them by hand. The instruction defines what it is measuring as the entire crop in the field, down to the last grain—the crop that could be gathered if there were no losses or theft at harvest and threshing at

all. And it states its own purpose: to forestall attempts to conceal from the state the real size of the harvest (Zelenin, 1970, pp. 465–466).

This is the asymmetry the rest of the paper measures. The commission's number was not an estimate of the grain a farm held. It was an estimate of the grain a field had carried, with every subsequent loss assumed away. Deductions for harvest losses were written into the instructions and, in practice, were nominal: at the June 1934 Central Committee plenum the commission's own deputy chairman reported that losses in 1933 had averaged 1.5–2 centners a hectare, a quarter to a third of what had grown. Table 1 sets the two numbers side by side for the same collective farms in the same years. The farms' own year-end accounts, collected in the dynamic surveys this paper uses, give an all-Union grain yield of 6.7 centners per hectare in 1933; the commission determined 8.8. In Ukraine the gap is wider: 7.8 against 11.2, and in every Ukrainian oblast the commission's figure exceeds the farms' own by between a fifth and a half—Kyiv 10.6 against 7.1, Dnipropetrovsk 12.7 against 8.3.

Two consequences follow, and both are the subject of later sections. First, a base measured on the standing crop is one an official can verify by walking a field, which is what made it enforceable; the wedge between it and the grain that reached the barn is the harvest loss, which is proportionally largest in a bad year. Second, the commission's figure is the number the fee of Article 12 was a percentage of. Zelenin puts the link plainly: for collective farms the use of the biological yield meant a higher payment for the station's work, because the payment in kind came to be computed on the commission's data (Zelenin, 1970, p. 466). Section 5 asks whether, in the one region where the statute and the realised payment can be observed on the same farm, the 1933 fee in fact tracked the district assessment or the farm's own crop.

Decree No. 390 of 17 February 1934 replaced the flat percentage with a schedule in kilograms per hectare graduated by yield band, the band fixed for a whole district; threshing remained at 8 per cent of the grain threshed and the combine at 11 (Sovnarkom SSSR, 1934). The two decrees are not the same instrument and are never pooled in what follows. No. 346 prices the work as a fraction of a farm's own crop; No. 390 prices most of it as a quantity per hectare read off a district band.

Three bases, and the year the two arms crossed. Every claim in this system was written on one of three things: the work the station had performed, the output the farm had grown, or the land the farm had sown. Table 2 records which base each arm used in each year, and how it was paid. The delivery moved from output to land in January 1933 and stayed there. The station's fee moved from work to output in February 1933, and from output to land in February 1934. The two arms therefore crossed: in the same weeks in which the delivery ceased to be a share and became a quantity, the fee ceased to be a price and became a share. By 1934 both claims were quantities per

Table 1: Barn and assessed grain yields.

	Survey (barn)		Commission (standing crop)		Ratio	
	1933	1934	1933	1934	1933	1934
USSR	6.7	6.6	8.8	8.5	1.31	1.29
RSFSR	6.3	7.1	8.3	9.1	1.32	1.28
Ukrainian SSR	7.8	4.8	11.2	6.0	1.44	1.25
Byelorussian SSR	6.6	7.0	9.1	9.2	1.38	1.31
Sverdlovsk oblast ^a	5.5	8.3	7.1	9.2	—	—
<i>Ukrainian oblasts, 1933 (the commission column is not printed by oblast for 1934):</i>						
Kyiv	7.1	5.9	10.6	—	1.49	—
Kharkiv	7.3	5.9	10.7	—	1.47	—
Dnipropetrovsk	8.3	3.3	12.7	—	1.53	—
Odesa	8.4	3.6	12.6	—	1.50	—
Donetsk	6.1	3.7	9.3	—	1.52	—
Chernihiv	6.2	6.2	8.8	—	1.42	—
Vinnytsia	9.6	6.8	11.7	—	1.22	—

Notes. Grain yield on collective farms, centners per sown hectare. The survey columns are the dynamic (*dinamika*) surveys of collective farms, RGAE f. 1562 *op.* 78, and report the grain the farms entered in their own year-end accounts after threshing. The commission columns are the Central State Yield Commission's determination, measured on the standing crop before harvest. The ratio is the commission figure divided by the survey figure.

^a Sverdlovsk shares a single commission cell with Chelyabinsk in the printed table, so no ratio is formed for it. It is shown because the farm audit used in Sections 5 is drawn from it.

Source: I. E. Zelenin, "Osnovnye pokazateli sel'skokhoziaistvennogo proizvodstva v 1928–1935 gg.," Table 1 (Zelenin, 1970).

hectare, set administratively, and held by two agencies with separate accounts and no mechanism for netting one against the other. The system arrived at that configuration through two decisions taken a year apart, by different bodies, for different stated reasons.

The design intent. The intent behind the fee's structure is stated directly in the contemporary record. In a memorandum of 1 February 1933, four days before the decree, the head of grain procurement M. A. Chernov rejected a scheme of payment for work actually performed, on the ground that it would let a farm served by a station hand over *less* grain per hectare than an unserved one, and asked instead for a fee structure under which a served farm surrendered at least two puds per hectare *more*. He proposed to enforce it by concentrating all threshing machinery in the stations and levying a compulsory threshing charge "even when the MTS tractors do no work" (Chernov, 1933).

In January 1933 the centre also installed political departments (*politotdely*) inside the machine-tractor stations; their mandate was the farms' obligations to the state rather than the stations'

Table 2: The base of each claim, 1932–1934.

Year	Obligatory delivery		MTS fee	
	Base	Paid in	Base	Paid in
Through 1932	output (share)	grain	work performed (price)	money
1933	land (c/ha)	grain	output (share)	grain
1934	land (c/ha)	grain	land (kg/ha by band)	grain

Notes. Delivery: Sovnarkom SSSR (1930) through 1932; decree No. 78 of 27 January 1933 (Sovnarkom SSSR, 1933b); decree No. 389 of 1934. MTS fee: fixed rates per unit of work in cash through 1932; decree No. 346 of 5 February 1933, art. 9 (Sovnarkom SSSR, 1933a); decree No. 390 of 17 February 1934 (Sovnarkom SSSR, 1934). Under No. 390 threshing (8 per cent of the grain threshed) and the combine (11 per cent) remain proportional; plowing, mowing and sowing become quantities per hectare. “Share” denotes a claim proportional to realised output; “c/ha” a fixed quantity of centners per sown hectare.

conduct, they were abolished in November 1934, and no politotdel-level data survive to test what they did (Online Appendix B).

What the two claims have in common. Neither rule refers to nationality. Both are functions of agricultural traits: sown area, crop, the region’s classification as grain-producing or consuming, the district’s assessed yield, and whether a farm was served by a station. That is what makes the apparatus measurable at all. It is also, as Sections 5 and 7 discuss, what makes an ethnic coefficient estimated without those traits difficult to interpret.

3 Data

This paper uses three bodies of data sitting at different levels: a sample survey of collective farms reported by region, the same survey’s primary forms for 272 farms in one oblast, and district handbooks for Ukraine, the Urals and the Lower Volga. Tables 3 and 4 list them.

3.1 Sources

The regional estimates come from the dynamic (*dinamika*) surveys of the economic and financial condition of collective farms, run by the Central Directorate of National-Economic Accounting (TsUNKhU) of Gosplan. Each round worked up the farms’ own annual reports for a pair of adjacent years and tabulated the results by region and by whether the farm was served by a machine-tractor station. The volumes are restricted typescripts and were never published. Three rounds survive for the period used here, covering 1930–32, 1932–33 and 1933–34.

Table 3: Sources: the dynamic surveys.

Source	Reference	Unit of observation	Years	Coverage
Dynamics of collective farms, from the annual reports	RGAE f. 1562, <i>op.</i> 76, d. 158; compiled 1934	region × 3 service groups	1930–32	23 regions, 138 rows
Dynamics of the economic condition, vol. I, oblast results	RGAE f. 1562, <i>op.</i> 77, d. 70	region, service groups pooled	1932–33	23 regions
— vol. II, group results	RGAE f. 1562, <i>op.</i> 77, d. 75	region × served / unserved	1932–33	12,707 farms; 80 cells used
Dynamics of the economic and financial condition	RGAE f. 1562, <i>op.</i> 78, d. 182; surveyed May–June 1935	region × 4 service groups	1933–34	39 regions ^a
— its primary forms, Sverdlovsk oblast	GASO f. 1812, <i>op.</i> 2, dd. 154–159 ^b	collective farm	1933–34	272 farms, 544 farm-years

Notes. The surveys were run by TsUNKhU of Gosplan; the volumes are restricted typescripts and were never published.

^a Not used for estimation. It is the source of the survey column of Table 1, and its field dates are what date the survey design described in Section 3.2.

^b GASO is the State Archive of Sverdlovsk Oblast. Men'shikova (2001) records the primary forms of this survey for 311 collective farms in 44 rayons of the oblast at this signature, together with the oblast's own working tabulations. The workbooks used here are transcriptions of such forms.

The farm-level data are the same survey's primary forms, transcribed in full for one oblast; Section 3.4 describes them. The district data are three statistical handbooks—one Ukrainian, one Ural, one for the Lower Volga—and the replication archive of Markevich et al. (2025), which supplies rayon mortality and the Ukrainian population share of the 1926 census; Section 3.5 describes those.

3.2 When the harvest was measured, and by whom

A collective farm drew up its annual report after the close of the economic year, from its own books. The yield in that report is the *barn* yield: the survey's own column heading reads “grain harvest (barn collection) per hectare sown”, and the accompanying tables record the crop as it was actually disposed of—to the state, to the seed and fodder funds, to the machine-tractor station, and to the members. This is what distinguishes the reports from the statistics published at the time, which carried the biological yield or something close to it (Men'shikova, 2001; Zelenin, 1970).

The surveys were compiled later still, and the 1933–34 round is documented step by step. TsUNKhU formed a standing panel of collective farms in the spring of 1935, intending to reach 17,500 of them, about 8 per cent. District and precinct inspectors were briefed on 12–15 May 1935 and visited the farms during May and June; the completed forms were due at the oblast office by 15 June, were checked there by 15 July, and the tabulation was finished by 1 September 1935. The

Table 4: Sources: district handbooks and the replication archive.

Source	Unit of observation	Years	Coverage
Handbook of district indicators, Ukrainian SSR ^a	rayon	1930–33	391 rayons
Ural handbooks and the 1934 oblast plan ^b	rayon	1926–34	58 rayons
Statistical handbook of the Lower Volga ^c	rayon	1929–32	67–70 rayons
Replication archive, Markevich et al. ^d	rayon; province	1928, 1933	1,907 rayons, 1,688 used ^e

^a Upravlinnia spravamy RNK USRR (1933), digitized in Harvard Ukrainian Research Institute (2023).

^b Ural'skoe oblastnoe statisticheskoe upravlenie (1930); Ural'skoe upravlenie narodno-khoziaistvennogo ucheta (1933); Sverdlovskaiia oblastnaia planovaia komissiiia (1934).

^c Nizhne-Volzhskaiia kraevaia planovaia komissiiia (1934).

^d Markevich et al. (2025).

^e A first difference in mortality requires a death rate in both years, which 1,688 of the 1,907 rayons have; 133 of the 137 Ural rayons.

stated purpose of the exercise was to follow the condition of collective farms from year to year *and to verify that their annual reports had been drawn up correctly*, and the inspector filled the form from the farm's own primary documents (Men'shikova, 2001). The earlier rounds were compiled on the same principle and with the same lag: the 1930–32 volume carries 1934 on its title page.

Two things follow, and both bound what this paper can claim. First, the survey measures grain that existed. It was compiled after the crop had been cut, threshed and taken into store, by inspectors sent to check the farm's books against its primary documents, which is why it is used here to measure what farms were left with and not to measure what they grew in the field. Second, it is not the number the state's claims were computed on. The Central State Yield Commission of Section 2 measured the standing crop before the harvest, because a claim has to be written before the grain moves and because the commission did not trust what farms would report afterwards. Table 1 therefore sets two different quantities side by side—measured months apart, by different bodies, for different purposes—and not two estimates of the same one. What separates them is everything that happened to the crop between the field and the barn.

The timing also settles the direction in which the two can be related. The claims of 1933 were computed on a figure produced before that year's harvest; the survey that measures the residue was compiled a year or more afterwards, by a statistical arm of the same agency, and cannot have entered the assessment. It follows that the survey cannot be used to audit the commission's arithmetic, and nothing below attempts it. What it can do is show what remained once claims written on the commission's figure had been met.

3.3 The sample, and the two groups

These are surveys of farms that filed. The 1930 model statute required a collective farm to draw up an annual report, and compliance was partial: 33 per cent of farms filed for 1930, 26.5 for 1931 and 40 for 1932 (Men'shikova, 2001). The 1932–33 round drew 12,707 farms, 7.5 per cent of all collective farms in the regions it covers—7.0 per cent in the RSFSR, 12.1 in Ukraine outside the Moldavian ASSR, 5.0 in the Belorussian SSR—and matched the same farms across the two years, so its tables are a two-period panel of identical units rather than two cross-sections. The sampling fraction is printed region by region and varies about six-fold, from 4.0 per cent in Western oblast to 24.8 in Stalingrad krai and 23.1 in Azov–Black Sea: the grain-surplus kraises are over-sampled relative to the northern consuming regions. Nothing below aggregates regions into a national figure.

The 1932–33 volume used here reports every region twice, once for farms served by a station and once for farms not served, and it is the only volume in which the payment in kind to the stations appears as a line of its own. That line is blank throughout 1932 and stands at 11.6 per cent of the gross harvest in Ukraine, 10.1 in the RSFSR and 9.2 in the Belorussian SSR in 1933. On all 92 rows of the distribution table the obligatory delivery, seed-loan repayment, payment in kind, collective funds and distribution to members sum to 100.00 per cent of the gross harvest (s.d. 0.08; range 99.7–100.6), so the shares are mutually exhausting: one cannot rise unless another falls.

Three features of the grouping bear directly on the design and are set out here rather than in a footnote, because each shapes how the estimates in Section 5 should be read.

(i) *The two cells are very unequal in size, and the smallest unserved cells are Ukrainian.* In 1933 the unserved cell ranges from 21 collective farms (Odesa) to 852 (Moscow); the four smallest are Odesa (21), Donetsk (38), Vinnytsia (47) and Dnipropetrovsk (59), all Ukrainian, against served cells of 445, 101, 299 and 302. Any test that asks whether the served gap *differs* in Ukraine is therefore identified where the Ukrainian control group is thinnest. Section 7 reports what that costs in power.

(ii) *Served farms are larger.* In 19 of 20 regions the served group has more sown area per farm, median ratio 1.53 (Vinnytsia 2.41, Kyiv 1.87). Level differences between the groups are accordingly part scale and part service, which is why the estimates below are read off the *change* at the rule change rather than off the level. The size gap is not constant either: between 1932 and 1933 unserved sown area grew 18.4% against 11.1% for served, so the gap narrowed in 17 of 20 regions. Section 5 shows the difference-in-differences is unmoved by controlling for that differential growth.

(iii) *Group membership is not fixed across the rule change.* Table 1 of the 1932–33 survey splits the 1933 served group into farms served *only* from 1933 and farms served in 1932 or earlier: 991 of 5,310, or 18.7%, of the 1933 served group had been in the control group a year before

(Ukraine 41.6%). In the cash-era 1930–32 panel the drift is larger still—the median served share of households runs from 13.3% in 1930 to 38.5% in 1932. The direction of the resulting bias is knowable: the entrants performed markedly less station work in 1933 than the incumbents (45.5 against 60.4 plow-equivalents per 100 hectares), so if the fee scales with work—which Section 5 shows it does—the 1933 served mean is diluted toward the control group and the estimated jump is *attenuated*. Composition makes the result conservative, not fragile. It is nevertheless a reason to read the difference-in-differences as a lower bound on the effect of the rule on a farm that was served throughout.

3.4 The farm audit

The published survey stops at the region. For one oblast its primary forms survive, and they carry both years of the pair on the same document, section by section: the farm’s population, labour-days and their distribution, sown area and gross harvest by crop, the distribution of that harvest, the work done for the farm by the machine-tractor station, livestock, seed, and the farm’s finances. I use a complete transcription of 272 such forms from Sverdlovsk oblast for 1933 and 1934—544 farm-years, geo-referenced. For each farm they record the gross harvest, the obligatory delivery, the state purchase (*khlebozakup*), the bill in kind from the station, the seed, insurance and fodder funds, the distribution to members, the physical volume of station plowing, mowing and threshing, and the farm’s population in each year. Service by a station is observed directly, which gives three groups: never served ($n = 111$), served from 1934 only ($n = 20$), and served in both years ($n = 71$).

Seven flag variables mark unusable farm-years and are applied before every estimate; the station arm additionally requires the form’s machine-tractor section to be usable, and drops the 48 farms that report a positive bill in kind with all five work shares blank. The audit is one oblast. It identifies comparisons between farms and within districts, and says nothing about differences between regions.

3.5 The district handbooks

Ukraine. The rayon layer of the MAPA project digitizes a handbook of district economic indicators published in eight volumes, one per oblast, by the administration of the Ukrainian government (Upravlinnia spravamy RNK USRR, 1933; Harvard Ukrainian Research Institute, 2023). It carries, by district, planned and actual grain procurement for three campaigns with a fulfilment rate for each, the collectivization rate in October–December 1932, sector and crop shares of the sown area, the distance from the district centre to a railway station, and estimates of direct population losses in 1933 from the demographic reconstruction of Wolowyna et al. (2016). Three properties of it govern its use here. There is no harvest, output or yield field anywhere in the atlas, so procurement as a

share of what a district produced cannot be formed at rayon level; every procurement variable is per rural inhabitant, and procurement per head was largest where output was largest, which makes it a measure of a district's capacity rather than of the burden it bore—the rayons of Odesa record about 384 tons per 1,000 rural inhabitants against 45 in Kyiv. Fulfilment is measured against a plan that was cut during the campaign, and cut by different amounts in different oblasts, so the cut and the fulfilment rate are reported together wherever either appears (Section 4). And the loss estimates are a reconstruction rather than an archival series, with the rayon figures rescaled to sum to oblast totals, so their between-oblast variation is better identified than their within-oblast variation.

The Urals. No rayon grain target survives for the Urals; what the sources print is sown area. I use a 58-rayon panel of total sown area on the 1934 Sverdlovsk frame for 1926, 1930, 1931 and 1932, with the 1934 sowing plan as a fifth column, built from the 1930 agricultural census and district handbook, the 1931–32 statistical handbook, and the oblast planning volume for 1934 (Ural'skoe oblastnoe statisticheskoe upravlenie, 1930; Ural'skoe upravlenie narodno-khoziaistvennogo ucheta, 1933; Sverdlovskaiia oblastnaia planovaia komissiiia, 1934). Rural households counted in the December 1926 census, before collectivization, give a predetermined measure of district size. Two construction choices matter. The 1934 column is a plan and not a realization, and it is annualized: the published table is the spring sowing plan, to which a winter component estimated from the volume's own oblast controls has been added. And the panel uses each rayon's total sown area rather than a sector split, because between 1930 and 1932 the individual sector fell from 1,296 to 304 thousand hectares while the collective sector rose from 497 to 1,903: a sector series in these years measures collectivization, not the size of the base a per-hectare claim was levied on.

The Lower Volga. The Lower Volga handbook publishes neither a target nor a procurement series, only realized sown area by rayon—1929–30 and 1930–31 in one table and a 1932 cross-section in another, 67 to 70 rayons depending on the year (Nizhne-Volzhskaiia kraevaia planovaia komissiiia, 1934). It enters only as a second region in which the persistence of the base can be checked.

3.6 What is observed at which level

Table 5 states, for each object the paper reasons about, the level at which it can be observed at all; Online Appendix A describes every variable, its construction, and the level and number of units at which it exists. Three of the table's rows do most of the work.

What a farm was left with after the state's claims is observed at the farm, in one oblast, and at the region, for twenty regions. It is not observed at the rayon anywhere, in any region, in any year, because no rayon-level harvest exists to net the claims out of. Ethnic composition and mortality,

Table 5: Levels of observation.

Object	Observed at	Units
The harvest and its distribution	the farm (one oblast); the region $\times$ service group	544; 80
What the farm was left with	the farm (one oblast); the region	544; 18–20
Mortality	the rayon; the region, by aggregation	1,688; 19
Ukrainian population share	the rayon	1,688
Collectivization	the rayon	391
Grain procurement	the rayon, all sectors, per rural head	391
Gross harvest or yield, at the rayon	<i>unobserved</i>	—

Notes. The last row is the binding constraint: no source prints a rayon-level harvest or yield in any region or year. Where a rayon-level residual is needed, Section 6 uses the two constructed measures described in Online Appendix A and labels them; neither carries a claim of its own.

by contrast, are observed at the rayon, and at no other level with more than twenty units. The two therefore never meet on equal footing: in a rayon-level comparison the ethnic variable is a measurement and the residual has to be imputed, while in a region-level comparison the residual is measured and there are eighteen observations. That is a statement about what these data can identify, and it applies to any use of them, including mine.

3.7 Consistency checks

The survey prints five welfare quantities separately—grain per labour-day, labour-days per worker, the share of the gross harvest distributed to members, gross grain per worker, and grain per soul—and an accounting identity links them, so each link can be tested rather than assumed. On the 80 region-group-years the identity closes to a median deviation of between 0.24 and 0.43 per cent depending on the route: the five are not independent measurements but one measurement decomposed five ways, and are treated as such. Two independent record systems agree with the survey where they overlap. Across Ukraine’s seven oblasts the district handbook’s procurement per 1,000 rural inhabitants ranks the oblasts as the survey’s delivery measures do (rank correlations +0.821 against delivery per hectare and +0.964 against the delivered share), and the household budget survey of collective farmers, which records what a household received from its kolkhoz, matches the farm-side distribution in level for the seven regions the two sources share (median ratios 0.92 and 0.97 against calendar 1933 and 1934) and in the ordering of the famine year (rank correlation +0.86). One quantity disagrees: fulfilment of the procurement plan correlates -0.75 with the share of the harvest surrendered, because fulfilment is a ratio to a plan that had been cut, and cut furthest where capacity was smallest—the reason no measure of burden in this paper is a fulfilment rate. Online Appendix A reports each check in full.

4 The Assignment and Rigidity of the Quota

Between 1929 and 1934 the figure a district owed was arrived at in two quite different ways. Until 1933 it was the outcome of an apportionment: a national total divided downward, settled at every step by negotiation, and revisable in both directions while the campaign ran. From 1933 it was a norm published for the district before the spring sowing, computed from a rate the region was obliged to fix and forbidden to alter. The two periods pose different questions—under an apportionment, what the number was carried forward from; under a published norm, at what level it was allowed to differ. Historians have reconstructed the apportionment from the archives of the procurement campaigns (Il'inykh, 2015; Kondrashin, 2015; Davies, 1980); the district plan enters the quantitative literature on the famine as a regressor, and what it was a function of has not been estimated. A target set on what the unit delivered last year is the ratchet of the literature on planned economies (Berliner, 1957; Weitzman, 1980; Freixas et al., 1985).

A claim settled before the harvest could not respond to it. What followed is taken in order: the harvest's movement against the standing claim, the terms into which the residual it left decomposes, the reward rate and the surplus region by region, and the two forms in which the same take could have been written.

4.1 The apportionment

The annual grain-procurement plan was a single national figure, and it reached a collective farm by being divided. The centre fixed the total and apportioned it to regions, regions to districts, districts to village soviets, and village soviets to farms; this downward division of a quantity fixed at the top is *razverstka*. What a farm surrendered under it followed the task it had been given and not the harvest it had grown (Il'inykh, 2015, pp. 364–366). The criterion by which the centre divided the 1930 total among regions was the growth of sown area, and the tasks were issued against the areas districts were planned to sow rather than the areas they sowed (Kondrashin, 2015, pp. 372–373).

The Urals reckoned their own task in the form the law would later require. In 1932 the two oblast secretaries reported a task of 75 million puds—48 from collective and individual farms, 9 from state farms, and 18 in repayment of the *seed loan*, the grain the state advanced for sowing and recovered in kind after the harvest—and put procurement and seed loan together at 2.1 centners per sown hectare, or 36.4 per cent of the gross harvest, ranging by zone from 30.8 per cent in the northern Trans-Urals to 47.5 in the south (Baranov et al., 2009, doc. 41).¹ The seed loan sat inside the task rather than beside it, and the claim was already per hectare and already differentiated within the oblast a year before either feature entered the law.

¹With the milling toll, a levy in kind on grain brought to the mills, the task came to 80 million puds, against an oblast yield put at 5.7 centners a hectare. A pud is 16.4 kilograms.

The *counter-plan* raised a task that had already been apportioned. It took the form of a pledge from below to exceed the figure received, and was in fact decided above and apportioned downward like the plan itself; district authorities also used it to move an unmet task from one farm onto another (Il'inykh, 2015, pp. 366–367). Regional and district authorities padded what they passed on, handing down more than they had received to insure themselves against failing to fulfil it. A farm that harvested well and completed its task was given a supplementary one at once, and could end the year with less grain for its members than a farm whose fields had failed (Il'inykh and Simonov, 2021, p. 11).

A task could also be lowered, but not at the speed at which it was raised. The Politburo cut the Ukrainian republic's figure by 40 million puds on 17 August 1932, as an exception for the districts “which especially suffered”; by a further 70 million on 30 October; and again on 12 January 1933. Each decision was to go unpublished, and the January cut was withheld from the districts, which kept the plans they already held while the reduced republic figure was collected in full (Davies and Wheatcroft, 2004, pp. 184–185, 201). Of the six Ukrainian oblasts the dynamic surveys cover, Kyiv had the worst of that harvest: 4.7 centners of grain per worker, and 0.70 a person distributed to members against the 1.5 a year's subsistence takes (Section 6). It also had the deepest cut of the seven, to 0.30 of its 1930–31 quota per thousand rural inhabitants. Its 76 districts then delivered 101.0 per cent of the reduced quota, against 89.4 across all 386 Ukrainian districts.

The law of 19 January 1933 ended the chain at the district, the level at which the binding norm was now written (Section 2). A decree of 20 June made the issue of a counter-plan a criminal offence, on the stated ground that a farm should be able to reckon its income in advance (Sovnarkom SSSR and TsK VKP(b), 1933a). Upward revision was removed; downward revision was not.

4.2 The assignment rule

A district's grain quota for one campaign stood close to what that district had delivered in the campaign before. Let T_r be the 1932–33 procurement plan assigned to district r , $A_{r,t-k}$ its realized procurement k campaigns earlier, and S_r its rural population in 1930:

$$\ln T_r = \alpha + \beta_1 \ln A_{r,t-1} + \beta_2 \ln A_{r,t-2} + \gamma \ln S_r + \varepsilon_r. \quad (4.1)$$

Entered alone, β_1 is 0.912 (s.e. 0.027) with $R^2 = 0.709$ on the 384 Ukrainian districts for which both quantities survive. The dependent variable is the plan; the regressor is realized procurement. The plan tracked the realization and not the previous plan. The two are nearly collinear across districts (correlations of 0.995 and 0.984 in the two transitions), so only the gain in fit separates them: on 373 districts, adding the previous plan to the previous realization raises R^2 by 0.0001 and by 0.0020 in the two transitions, while adding the previous realization to the previous plan raises

Table 6: The assignment of the district quota, 1932–33.

Dependent variable: ln planned grain procurement, 1932–33	(1)	(2)	(3)	(4)
ln procurement 1931–32 (β_1)	0.912*** (0.027)		0.582*** (0.067)	0.677*** (0.064)
ln procurement 1930–31 (β_2)			0.340*** (0.061)	0.306*** (0.057)
ln rural population, 1930 (γ)		0.408*** (0.114)		−0.307*** (0.056)
R^2	0.709	0.036	0.745	0.761
Districts	384	385	373	372
$p: \beta_1 = \beta_2$			0.054	0.002
$p: \beta_1 + \beta_2 = 1$			0.002	0.523
$p: \beta_1 = 0$				< 0.001

Notes. OLS estimates of equation (4.1); all variables in logs; heteroskedasticity-robust standard errors in parentheses. Column (2) enters rural population alone. The three tests at the foot are what separate a quota carried forward from past deliveries from one assessed against district capacity: under the latter the two lags are interchangeable and a direct measure of size makes them redundant. Both are rejected. Proportional carry-forward implies $\beta_1 + \beta_2 = 1$, which is rejected before population is controlled for and not rejected after. The corresponding estimates for the Urals and the Lower Volga, where no district target was published and the dependent variable is realized sown area, are reported in the text. *Source:* the Ukrainian district handbook, Table 4. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

it by 0.0066 and by 0.0370.

Two campaigns of past deliveries do not enter the quota equally. With both lags and rural population, on 372 districts, $\beta_1 = 0.677$ (s.e. 0.064) and $\beta_2 = 0.306$ (s.e. 0.057), with $R^2 = 0.761$ (Table 6). Equality of the two is rejected at $p = 0.002$, and with oblast fixed effects at $p < 0.001$. A quota assessed against what a district could produce would not distinguish them, since one past campaign stands in for productive capacity as well as another; a quota carried forward weights the most recent. The centre said something similar in July 1932, when it held that the 1931 plan had been apportioned to districts and collective farms mechanically and without regard to the position of each, and named Ukraine and the Urals (TsK VKP(b), 1932). It was assigning blame for a failed campaign, and is better read as a characterisation of the apportionment than as evidence about it.

The Kyiv oblast handbook for 1932 prints, district by district, arable land, households, draft animals, tractors and grain area. Matched to the procurement layer it yields 82 districts—the handbook covers the oblast as it stood before October 1932, which in the later frame is 55 districts in Kyiv, 22 in Chernihiv, 3 in Kharkiv and 2 in Dnipropetrovsk—of which 79 carry the whole vector. Arable land enters as scale and the rest per hectare, the levels being too nearly collinear to

separate. On the 1930–31 plan, the earliest on record, the capacity block alone reaches $R^2 = 0.640$ and is jointly significant ($F = 14.28$, $p < 0.001$). On the 1931–32 plan capacity alone reaches 0.558 against 0.869 for the lagged realization, and adding capacity to the lag raises R^2 by 0.008 ($F = 1.39$, $p = 0.237$) while adding the lag to capacity raises it by 0.319. On the 1932–33 plan the increments are 0.076 for capacity ($F = 3.88$, $p = 0.004$) and 0.103 for the lag. Capacity is in the number at the start of the chain and is displaced by the carry-forward once there is a chain to carry. The individual coefficients are not read here: draft animals per hectare enters negatively, which marks poor land rather than a rule.

The quota on record is the figure as it stood at the end of the campaign, and quotas were revised. The revisions answered to what a district had delivered: on 373 districts the cut between the 1930–31 and the 1932–33 campaigns loads on the district’s fulfilment ratio in 1931–32 at +1.370 (s.e. 0.211), and on the 1930–31 ratio, which precedes both later plans, at only +0.515 (s.e. 0.265). Harvests are persistent, so a district that delivered little in one year may have been cut in the next for the same underlying reason, and the lag coefficient would then be measuring the revision and not a rule.

Grain procured in 1929 precedes every plan in the layer. The 1930–31 plan—the earliest, for a campaign whose mean district fulfilment was 101.3 per cent and which therefore had little occasion to be cut—is a near unit-elastic function of what the district procured in 1929, at 0.970 (s.e. 0.050) with $R^2 = 0.70$ on 271 districts, and grain procured in 1929 alone explains 56 per cent of the quota assigned three campaigns later. The earlier transition, whose revision history is a different year’s harvest, gives the same answer, at 0.799 (s.e. 0.028) on 372 districts. Absorbing what makes a district persistently productive attenuates the carry-forward without removing it: wheat share, rural density, collectivization and distance to rail move β_1 from 0.677 to 0.595 on 310 districts, oblast fixed effects move it to 0.508, and both together to 0.403, while the contrast between the two lags is unaffected.

Two results in the pooled specification hold between oblasts and not within them. The two elasticities sum to unity once rural population is held fixed ($p = 0.523$), but with oblast fixed effects they sum to 0.663 and proportionality is rejected ($t = -9.46$), so proportional carry-forward is a feature of the comparison across oblasts. And the coefficient on rural population, -0.307 (s.e. 0.056) pooled, reverses to $+0.122$ (s.e. 0.059) inside oblasts. Densely settled districts received smaller quotas than the steppe did; within an oblast, a district with more people received a larger one.

Neither the Urals nor the Lower Volga published a district target, and what persists in those two regions is the base the per-hectare claim was levied on. A Ural district’s total sown area is a near unit-elastic function of its own previous level, 0.983 (s.e. 0.044) with $R^2 = 0.961$ on 56 districts, and unit elasticity is not rejected ($p = 0.70$); rural households counted in the December

1926 census, before collectivization, explain 63 per cent of 1932 sown area on their own but take a coefficient of -0.017 beside the lagged area. In the Lower Volga the elasticity is 1.076 (s.e. 0.034) with $R^2 = 0.930$ on 67 districts. A base carried forward in hectares takes no account of the people left to work them: across 43 agricultural Ural districts the rural population fell by a median 17 per cent between 1928 and 1933, and sown hectares per rural household rose from 3.85 in 1930 to 4.45 under the 1934 plan. The elasticity of the ratio of the 1934 plan to 1932 area with respect to a district's change in rural population is $+0.26$ (s.e. 0.24), so one cannot reject that the target ignored the shrinking labour force and one can reject that it tracked it.

4.3 The level at which the claim varied

Two rules governed the grain claim in these years, and Section 2 states both: until the 1932 harvest the claim was a share of gross output, set by the list a region stood on; from the law of 19 January 1933 it was a fixed quantity per sown hectare. The claim itself is observed at three levels—regions in 1931–33, the Ukrainian districts of the 1932–33 campaign, and the audited Ural farms of 1933 and 1934—and each level carries the test of the rule in force there.

Under the first rule the claim was a fraction of the crop, so the claim per hectare is the fraction times the yield. Write d_{it} for state grain delivery per sown hectare of region i in year t , in centners, and Y_{it} for its survey barn yield, estimated separately on each statutory list g :

$$d_{it} = \alpha_g + a_1^g Y_{it} + \varepsilon_{it}, \quad g \in \{\text{producing, consuming}\}. \quad (4.2)$$

The equation is in levels rather than logarithms because under proportionality the level slope is the share itself, the quantity the 1930 norms bound: one quarter to one third on the producing list, at most one eighth on the consuming. The consuming list sat at its statute: across the 17 consuming region-years of 1931–33 the mean share of harvest delivered is 0.112 and the slope is 0.084 (s.e. 0.065). The producing list was held above its statute: on 25 producing region-years the mean share is 0.503 in 1931, 0.453 in 1932 and 0.364 in 1933, over the written third in every year (Online Appendix B). The ceiling restrained the claim only where the claim was small. A share collected above its ceiling is also what an overstated base produces: the campaign worked from assessed rather than barn yields, the assessed figure ran roughly thirty per cent above the actual where the two can be compared (Table 1), and a statutory third of a crop overstated by that margin is 0.43 of the true one—close to what the producing list delivered.

The change of rule shows in the slope. Within the producing list the slope of delivery per hectare on yield falls from 0.531 (s.e. 0.049 , 6 regions) in 1931 to 0.283 (s.e. 0.202 , 11 regions) in 1932 and 0.131 (s.e. 0.055 , 8 regions) in 1933: in 1931 the slope equals the share, which is what a proportional levy implies, and by 1933 the claim has nearly stopped moving with the yield

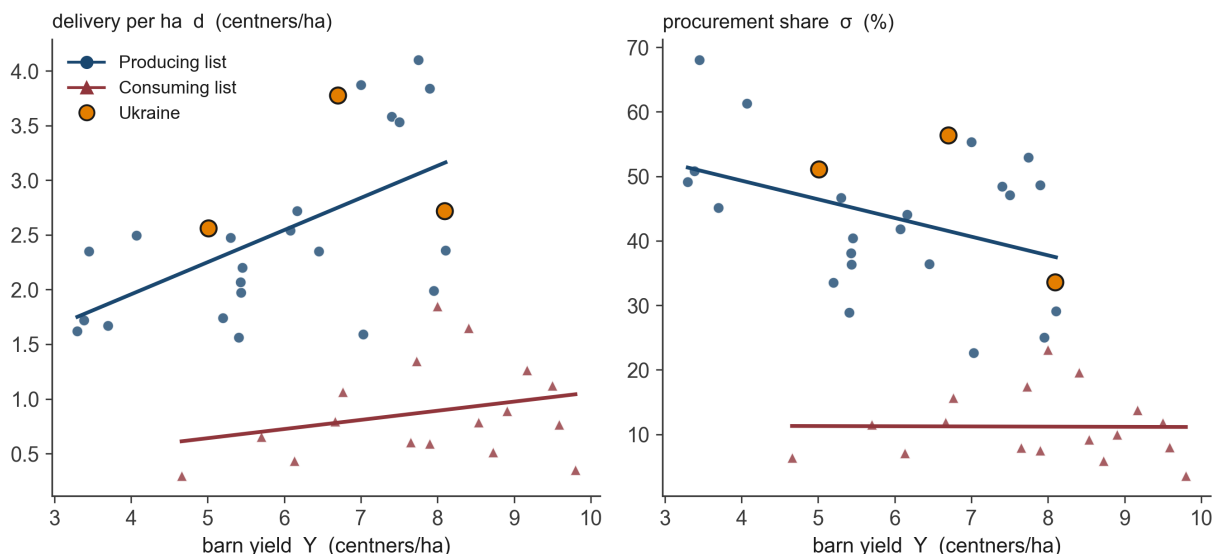


Figure 1: Grain delivery and barn yield by statutory list, 1931–1933.

Notes. Region-level outcomes against barn yield Y for grain-producing (navy circles) and consuming (maroon triangles) regions; the Ukraine unit is highlighted in orange (three points, 1931–33). Lines are the OLS fit within each group (Ukraine is included in the producing fit). *Left:* delivery per sown hectare d (slope a_1). *Right:* procurement share $\sigma = d/Y$. Producing regions sit on a higher per-hectare schedule, and Ukraine sits squarely on it—its high delivery is a producing-status position, not an ethnic outlier. Within each group delivery rises with yield while the share falls. Standard errors clustered by region. Sample: 1931–1933 ($N = 42$; 25 producing, 17 consuming region-years; the surveys carry no usable 1930 yield). *Data:* the dynamic surveys, Table 3, aggregated to the province frame of Markevich et al. (2025); d is delivery per sown hectare and Y is the survey barn yield of Section 3.2, not the biological figure.

while its level stands, which is what a fixed per-hectare norm looks like from region level. The by-year cells hold 4 to 11 regions, so the slopes are read as a pattern and not singly. Over the whole window, of a marginal centner of yield the claim on a producing region absorbed 0.29 (s.e. 0.085) and the region retained the rest.

Ukraine was taxed as a producing region. Its share of harvest delivered—0.564, 0.511 and 0.336 across the three years—runs beside the producing-list mean of 0.503, 0.453 and 0.364, and an indicator for Ukraine added to the producing-list equation takes 0.35 centners per hectare (s.e. 0.23). Figure 1 shows the same result graphically: the left panel plots delivery per hectare against barn yield with the fitted line of each list, the right panel plots the delivered share, and Ukraine’s three region-years lie on the producing-list line, not above it. At this level the claim on Ukraine is the claim on the list it stood on, a fact Section 7 returns to.

The district is the level at which the claim differed most. Across the 356 Ukrainian districts of the 1932–33 campaign the standard deviation of log procurement per rural soul within an oblast is 0.470, and the district at the 90th percentile of its oblast owed between 2.25 (Vinnytsia) and 5.51 (Donetsk) times the district at the 10th (Online Appendix B). Districts differ in the land their

households worked, and a claim reckoned per soul could simply reflect that endowment; it does not: across the 82 districts of the matched Kyiv frame the elasticity of the claim per rural soul with respect to sown area per household is 0.22 ($t = 0.8$), and on the 95 districts with a validated sown area the per-hectare spread equals the per-soul spread (0.471 against 0.470). By the assignment rule of Section 4.2 the differences were instead carried forward from the districts' own past deliveries. No district harvest survives in any Ukrainian source, so the share rule cannot be checked at the level at which the claim differed most.

Below the district the 1933 law wrote two rates, one for farms a machine-tractor station served and one for the rest. Within the same district a served farm's delivery per hectare stood 0.18 log points below an unserved one's in 1933 and 0.23 below in 1934—17 and 20 per cent—on 232 and 248 audited farm-years ($t = 8.2$ and 11.1): the lower rate the law set for farms that also owed the station its payment in kind, the second claim Section 5 follows. Measured at the cell the law defined, district by station status, the claim is close to uniform: 75.8 per cent of the variance of the log claim lies between cells in 1933 and 68.4 in 1934, the dispersion within a cell holds at 0.11 in both years, and inside a cell the claim is a third narrower than the harvest (ratios of standard deviations 0.63 and 0.68). The between-district share alone falls from 62.3 to 44.5 per cent, which is the two rates moving apart between the two laws and not farms inside one norm drifting apart.

The farm's own harvest is the last test, and the sharpest, because the two rules predict opposite ends: a share of the crop makes the elasticity of the claim in the farm's own harvest one, and a per-hectare norm makes it zero. Write D_{fc} for the delivery per sown hectare of farm f in norm cell c :

$$\ln D_{fc} = \alpha_c + \gamma \ln Y_{fc} + \varepsilon_{fc}, \quad \text{share rule: } \gamma = 1; \quad \text{norm: } \gamma = 0. \quad (4.3)$$

In 1933, on 232 farm-years over 75 cells, $\gamma = 0.162$ ($t = 3.61$), and both ends are rejected at $p < 0.001$: the first campaign under the law stands between the two rules, no longer the share and not yet the flat norm. In 1934, on 248 farm-years over 77 cells, $\gamma = -0.031$ ($t = -0.60$), the norm as written; with district effects alone the pair is 0.168 and -0.001 (Online Appendix B). The station changed the level of a farm's claim and not its slope: in 1933 the elasticity is statistically indistinguishable between the farms a station served and the rest (0.078 against 0.163; difference $t = 1.1$, district effects and the status wedge held, on the same 232 farm-years). The winter share of a farm's sowing, fixed in the autumn of 1932 before the law existed, does not carry the 1933 gradient either: holding it fixed moves γ from 0.168 to 0.169.

At every level, then, the claim followed the rule in force. While the rule was a share, regions delivered by list—the consuming list at its statutory eighth, the producing list past its written third—and the claim moved with the harvest. When the rule became a norm, the claim went flat in the harvest at the region and at the farm within a year of each other. What the change of rule left standing was the spread between districts.

4.4 The moving harvest

A claim fixed in advance of the harvest turns a production shock into a subsistence shock: when the crop falls and the state's take does not, the whole shortfall lands on what remains to the farm. The assignment rule of Section 4.2 set the claim in advance at every level; whether the harvest's movements in fact fell on the residual is a question about years rather than levels, and the survey panel observes the same regions through the two harvests on either side of the change of rule.

Between the 1932 and the 1933 harvests the crop moved and the claim did not. Across the 20 survey regions—13 Russian and Belorussian regions and the 7 Ukrainian oblasts—median barn yield rose from 5.42 to 7.48 centners per hectare: steeply in Ukraine, +32 to +83 per cent by oblast, and weakly or negatively across much of the RSFSR (Stalingrad −2, Tatar −9, Western −6 per cent). The median claim per hectare moved from 1.90 to 1.98 centners. The median share of the harvest taken therefore fell from 32.2 to 27.1 per cent, and it moved in opposite directions in the two halves of the panel: down where the harvest recovered, up where it did not. Figure 2 places the two years in the longer record, by statutory list. The delivered share follows the list a region stood on throughout 1930–1933, Ukraine moving with the producing list. The delivery per hectare fell between 1931 and 1932 and then held across the change of rule: Ukraine surrendered 2.6 centners a hectare from the failed 1932 harvest and 2.7 from the recovered 1933 one. The grain left to members per worker carries the incidence: in Ukraine it falls to 1.8 centners in 1931 and 1932, against 2.9 to 4.1 on the Russian producing list, and recovers only with the 1933 harvest.

The two rules of Section 4.3 predict opposite responses. A claim proportional to the realized crop holds the share fixed, so the change in the share is unrelated to the change in the yield; a claim fixed in advance leaves the share to move inversely with the harvest, with elasticity minus one. Regressing the proportional change of the share on the proportional change of the yield across the 20 regions gives -0.79 ($t = -3.3$, $R^2 = 0.38$): the claim behaved as a quantity fixed before the harvest was known. The result is not Ukraine's recovery alone—on the 13 regions outside Ukraine the elasticity is -1.50 ($t = -2.6$), the fixed-claim benchmark itself—and the claim per hectare shows the same thing directly, responding to the yield's change at -0.08 ($t = -0.3$) on the same 20 regions. Nor does the result depend on which of the two archival series supplies the 1932 base: with the dynamic surveys' retrospective 1932 figures in place of the annual reports', the elasticity is -0.56 ($t = -2.1$) on the same 20 regions. That the per-hectare claim carried over nearly unchanged across the change of rule is Section 4.2's result seen from the other side: the new norms were written close to the take they replaced. The comparison begins at 1932: for the earlier years the only source is the kolkhoz annual reports, the series whose inconsistency led the statistical administration to replace it with the dynamic surveys (Zelenin, 1968; Men'shikova, 2001).

What a fixed claim does to incidence follows arithmetically, and 1933 ran both directions of

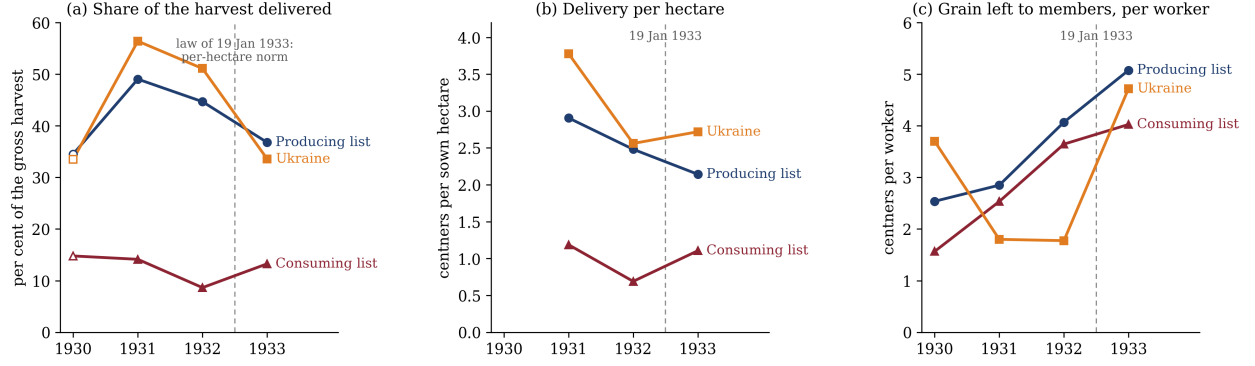


Figure 2: Delivery and the grain left, 1930–1933.

Notes. Unweighted means across the regions observed in each year, on the province frame: the RSFSR producing list (navy circles; 5–10 regions per year), the RSFSR consuming list (maroon triangles; 3–6 regions), and Ukraine (orange squares; one unit). *Panel (a):* the obligatory delivery as a share of the gross grain harvest. For 1931–33 the share divides the delivery per hectare by the survey barn yield, the region-level construction of Section 4.3; no source carries a barn yield for 1930, so the 1930 points (open markers) are the sources’ own printed shares, computed on the annual-report basis — the two bases are not comparable, and the 1930 level is shown for orientation only. *Panel (b):* the obligatory delivery per sown hectare, in centners; the panel begins in 1931 because the per-hectare figure also requires the barn yield. *Panel (c):* grain distributed to members per able-bodied worker, in centners, from the same volumes; the two survey rounds agree at their 1932 overlap to within 0.08 centners. Where both rounds report 1932, the 1932–33 survey is used, as in Section 4.3; the rounds differ on Ukraine’s 1932 delivered share (51.1 against 38.6 per cent — their farm samples differ), and the survey figure is shown. The dashed line marks the law of 19 January 1933, which replaced the share of the harvest with a per-hectare norm. The figure reports means and supports no estimate; the regressions of this section run on the 20-region survey panel. *Source:* the consolidated reports and the dynamic surveys, Table 3.

it at once. Ukraine’s recovering yields carried its share down—the farms kept the recovery. The non-recovering Russian regions surrendered a rising share of a flat harvest. No decision assigned either outcome: it is the norm of Section 4.3, held still while the harvest moved.

4.5 The composition of the residual

What the residual meant for a member is a quantity the survey prints from two directions at once. Let q be grain distributed per labour-day, in kilograms; t_d labour-days worked per able-bodied member; W/S workers per soul; σ the share of the gross grain harvest distributed to members; y the barn yield of grain, in centners per sown hectare; a socialised sown hectares per worker; and ϕ the grain share of the sown area. Grain distributed per soul, n , satisfies

$$n = \underbrace{\frac{1}{100} q t_d}_{\text{labour route}} \frac{W}{S} = \underbrace{\sigma \phi y a}_{\text{land route}} \frac{W}{S}, \quad (4.4)$$

and at energy densities K_g for grain and K_p for potato distributed at p kilograms per labour-day the caloric residual is

$$\text{kcal per soul per day} = \frac{W}{S} t_d (K_g q + K_p p) / 365. \quad (4.5)$$

The crop-mix term ϕ is not decoration: y is a grain yield while a is total sown area, so a land route written without it overstates the grain harvest by roughly a quarter at the grain shares the survey reports. Every term is printed independently, and the identity closes to within half a per cent (Section 3.7), so the terms can be read as a decomposition.

The identity makes the question left by Section 4.4 exact: when the residual differs across cells, which term differs with it? In logarithms both routes are additive, so the variance of the log caloric residual—equivalently of log grain per soul, the two differing by a constant—splits without remainder into one covariance share per term, $\text{Cov}(\ln x, \ln n) / \text{Var}(\ln n)$, and the shares sum to one by construction. On the 80 cells of the 1932–33 survey—20 regions, by served and unserved group, in each of the two years—the labour route gives 0.81 to the reward rate q , 0.19 to labour-days t_d , and 0.003 to workers per soul; the land route gives 0.41 to the yield, 0.28 to the distributed share σ , 0.23 to land per worker, and 0.08 to crop mix. Between a fed cell and a hungry one, the difference is what a labour-day paid, not how many were worked. On the land side, how the harvest was divided carries two-thirds of what the yield itself does, and more than the land a farm held per worker. The shares are accounting, not causes: they state where the variation sits, and correlated terms divide the credit between them. Nor is a low reward rate a statement about effort— q is grain to members over days worked, so a claim that leaves little to distribute makes q small by arithmetic.

These cells are as fine as the source permits, and three limits of its published form matter here. Each welfare variable is grouped separately—counts of farms by bins of one variable at a time—so the joint distribution is not observed: which farms stood low on the reward rate and on the yield at once cannot be read from the source, the decomposition runs on the eighty cell means, and what the unobserved coupling could change is bounded where the residual meets mortality (Section 6). The groupings count farms rather than people, and the farm-count mean exceeds the size-weighted mean in 57 of the 69 region-years where both are printed, so a share of farms below a line understates the share of members below it. And the cells are the 1932–33 dynamic survey’s own, compiled after the years they describe (Section 3.2) on the farms still in existence to report, so the worst-hit farms of the worst year may be missing from it.

4.6 The reward rate across regions

A labour-day could pay little for two reasons that a quotient does not distinguish. Grain distributed over days worked falls when the crop per worker fails—little sown land per worker, a poor yield,

or both—and it falls when the claim takes most of an ordinary crop and leaves a small share to divide. The two configurations end in the same figure but differ in which term of equation (4.4) moved: one farm grew too little, the other kept too little. The survey prints the terms region by region, so the two can be told apart.

The seven Ukrainian oblasts of 1932 divide between the two configurations (Table 7: region-level printed values of the dynamic survey, with the twelve Russian regions of the panel grouped by the statutory lists of Section 4.3). Kyiv and Chernihiv grew too little. They held consuming-belt land—1.5 and 1.9 sown hectares per worker, beside 1.6 across the Russian consuming list—at half the consuming list's yield; the delivery took the smallest share in the republic, 29.0 and 24.3 per cent of the harvest, and the share distributed to members, 27.0 and 33.0 per cent, was the largest; yet 78.6 and 66.7 per cent of their collective farms paid two kilograms per labour-day or less. The consuming list itself shows the same land carrying a liveable labour-day where the yield was double and the delivery under a tenth of the harvest. Kharkiv, Donetsk and Dnipropetrovsk kept too little: land per worker at the Russian producing list's level, yields no worse than Kyiv's, a delivery of between a half and two thirds of the harvest, members left 11–18 per cent, and the three highest headcounts in the republic, 84 to 91 per cent of farms at two kilograms or less. Odesa ran the same division on the most land per worker of the seven and posts the lowest headcount, 59.0 per cent: the claim cut less deep where the base per worker was largest.

Neither configuration is Ukrainian. On the Russian producing list the same claim met the failed Volga and Kuban harvests: Stalingrad delivered 68.0 per cent of a 3.5-centner crop and the Azov–Black Sea krai 61.3 per cent of 4.1, members in both were left about 13 per cent, and the Azov headcount below two kilograms, 88.7 per cent, stands within two points of Donetsk's, the worst in the survey. Vinnytsia, in turn, fits neither configuration: the best yield in the republic at 7.6 centners and a middling claim of 37.8 per cent—and still two thirds of its farms at or below two kilograms. The groupings beneath its means are wide, and a mean that describes the oblast describes few of its farms.

An average over the republic therefore mixes farms that grew too little with farms that kept too little: the pooled Ukrainian headcount lies between Chernihiv's and Donetsk's and corresponds to no oblast. The same heterogeneity returns at district level, where collectivization's association with mortality carries opposite signs across the Ukrainian oblasts (Section 7).

4.7 The surplus and the claims on it

Whether the farms below subsistence were put there by their fields or by the claims on them is a question the survey's own shares can settle. Each farm's printed yield, priced in calories per soul per day, is read on four successively smaller bases: the surplus, meaning the crop net of the seed

Table 7: The reward rate across regions, 1932.

	yield c/ha	sown ha per worker	delivery % of harvest	to members % of harvest	grain kg per labour-day	% of farms ≤ 2 kg
<i>Ukrainian oblasts</i>						
Chernihiv	4.7	1.9	24.3	33.0	1.84	66.7
Kyiv	4.1	1.5	29.0	27.0	1.54	78.6
Vinnitsia	7.6	1.4	37.8	25.2	1.88	66.0
Kharkiv	4.8	2.3	52.5	17.6	1.39	87.9
Donetsk	4.2	4.4	54.7	14.1	1.20	90.5
Odesa	5.4	4.7	63.3	13.8	2.07	59.0
Dnipropetrovsk	5.1	4.4	62.4	11.2	1.39	83.7
<i>RSFSR</i>						
Producing list, mean of 7	5.8	3.7	43.0	22.6	3.06	36.4
<i>of which</i> Stalingrad	3.5	7.2	68.0	13.1	2.08	60.1
<i>of which</i> Azov–Black Sea	4.1	4.4	61.3	13.6	1.25	88.7
Consuming list, mean of 5	8.3	1.6	9.8	44.0	3.63	26.2

Notes. Region-level values of the dynamic survey, 1932: the seven Ukrainian oblasts (ordered by the members' share) and the twelve Russian regions of the panel, grouped by the statutory delivery lists of Section 4.3; both named krajs belong to the producing list, and the list rows are unweighted means across regions. Yield is the printed barn yield of grain; delivery and to-members are the printed state grain delivery and grain distributed to members, per cent of the gross grain harvest. Sown hectares per worker combines the survey's two service-status cells (the survey prints no pooled 1932 figure). Grain per labour-day is the mean of the survey's grouping of collective farms, with the open top bin at 1.35 times its lower edge; where the survey prints its own ratio of aggregates the two agree to a median 4.4 per cent. *The last column is exact:* the bottom bin of the grouping is closed at two kilograms, so the headcount involves no interpolation. *Source:* Table 3.

and fodder the next sowing required; the surplus net of the obligatory delivery; net of every claim of the state; and the share finally distributed to members. The bases shrink in that order, so the share of farms below a subsistence line can only rise from column to column — and each rise names its taker: the delivery, the rest of the state's claims, the collective's own withheld funds.

Table 8 reports the share of collective farms below 1,600 kcal per soul per day, a working ration, on the four bases: 20 regions in both years, grain only, the Russian regions split by the statutory lists of Section 4.3. Each base reprices the farm's printed barn yield through the land route of equation (4.4) and shrinks it by the region's printed share of the gross harvest taken at that step, so all four columns read one distribution, and the headcounts are exact at the printed bins.

In the famine year the wedge belongs to the state exactly where the claim was heavy. Of the Ukrainian block's collective farms, 15.3 per cent were below the line on their surplus, 81.3 net of the obligatory delivery, 91.2 net of every state claim, and 95.6 on what was distributed: the state's claims contribute 76 of the 80 points of the rise, the delivery alone 66. The Russian producing list has the same shape at half the scale — 4.6 on the surplus, 41.5 after the state's claims, 47.7 distributed. On the consuming list the ordering reverses: the state's steps add 11 points, the collectives' own funds 19, and a fifth of the farms were below the line on their surplus

Table 8: Farms below subsistence, per cent, at each step of the division.

Block	Year	% of farms below 1,600 kcal per soul per day, counting the grain. . .				rise due to state claims (3)–(1)
		(1) grown, net of seed and fodder	(2) . . . and after the delivery	(3) . . . and after all state claims	(4) as actually distributed	
Ukraine (7)	1932	15.3	81.3	91.2	95.6	+75.9
	1933	3.6	13.5	19.9	33.2	+16.3
RSFSR producing list (7)	1932	4.6	37.3	41.5	47.7	+36.9
	1933	1.4	18.0	31.8	35.4	+30.4
RSFSR consuming list (5)	1932	22.0	32.1	33.3	51.9	+11.3
	1933	8.2	24.0	26.6	36.4	+18.4
BSSR	1932	30.5	30.5	30.5	76.0	+0.0
	1933	8.5	8.5	37.3	37.3	+28.8

Notes. Unweighted means across the regions of each block (regions in parentheses; the Russian regions grouped by the statutory delivery lists of Section 4.3); grain only. One distribution throughout: each farm's printed barn yield, priced at the region's printed grain share of sown area, sown hectares per worker and workers per soul, and scaled by the printed shares of the gross harvest — net of the seed and fodder funds the next sowing required (the *surplus*), then also of the obligatory delivery, then of the seed-loan return and the station fee (blank in 1932 by policy), down to the share distributed to members. The scalars shrink in that order, so the headcounts can only rise, and each rise names its taker: the *state-claims* column is (3) less (1), as the header states, and the rise attributable to the collective's own withheld funds is the remaining step, (4) less (3), readable from the columns. On the gross harvest before any reserve the 1932 headcounts are 11.7 (Ukraine), 1.6 (RSFSR) and 30.5 (BSSR) per cent, so nothing turns on how the reserve is treated. Headcounts are exact at the printed bins and unchanged to one decimal as the open top bin moves over [1.25, 2.00] times its lower edge. Cross-check: the survey's own reward grouping repriced through equation (4.5) (potato term zero) gives 90.3, 48.3, 44.1 and 67.0 down the four blocks in 1932 — within 0.6–9 points of the last column; in 1933 the two agree to 0.7–16 points. In half the region-years that construction coincides with the printed ≤ 2 kg headcount, the line falling in the same printed bin. The 1932 West Siberian delivery share is printed as 52.9, inconsistent with its own row (which then sums to 120.1) and with both external series; it is read as 32.9. Single-region steps are coarse (a shifted line lands inside a printed bin); the block means are the readable object. *Source:* Table 3.

to begin with. Belorussia is the limiting case: a consuming-list republic delivering 6.4 per cent of its harvest, its state steps are zero, and its wedge of 46 points is entirely the work of its collectives' funds on a failed crop. In 1933 the same arithmetic runs in reverse: under a claim that stood while the harvest moved (Section 4.4), the Ukrainian count as distributed falls from 95.6 to 33.2, while the two Russian lists fall by twelve and fifteen points. Kyiv remains the exception Section 4.6 classified: 43.1 per cent of its farms were below the line on their own surplus, the survey's highest, before any claim was taken.

4.8 Two rules for the same take

The form of the claim was itself a decision taken under the information problem. A claim written as a share of the realized crop taxes a quantity the extractor cannot verify — the crop was concealable — while a claim written on the hectare taxes what any inspector can measure and guarantees the regional total in advance; the two pure rules can both be computed on the printed distributions,

and for the same grain taken from a region they choose different victims, because their residuals cross at the regional mean: the fixed claim takes less than the share from every farm above it and more from every farm below it. In the Ukrainian block of 1932, where the take was nearly half the harvest, the share rule pushes 91.2 per cent of collective farms below the working ration of 1,600 kcal per soul per day, while the rigid rule returns the strong farms above the line (69.9 per cent below) but confiscates the entire surplus of 23.5 per cent of farms; at the 804 kcal that Meng et al. (2015) set as the calories for staying alive, the fixed claim is worse on both counts (40.4 against 33.0 per cent in Ukraine, 33.8 against 11.9 on the Russian producing list). Where the take was small the two rules coincide, the check that the difference measures the instrument and not the arithmetic. Concealment is why the share rule was infeasible, the choice Weitzman (1974) formalized: a state that could not verify the base and would not carry the risk wrote the quantity, and its price, at the same total, was the bottom of the distribution — the deep shortfall on which the mortality evidence of Section 6 loads. Online Appendix B reports the construction, its limits, and the full comparison for all twenty regions in both years.

5 The Machine-Tractor Station Fee

A machine-tractor station held the tractors and harvest machinery of a district; the collective farms of its zone worked their fields with its machines and paid for the work, under a contract in name only — its text fixed by decree, its fee enforceable without court (Section 2). Until 1933 the farms paid in money, at fixed rates per unit of work; in February 1933 the state turned the payment into grain, a share of the harvest, and from 1934 a fixed weight per hectare (Section 2). The design was stated in advance: days before the decree, Chernov's memorandum rejected payment for the work actually performed and demanded that a served farm surrender more grain than its unserved neighbour (Section 2).

In the famine years the station's claim stood on much of the collective harvest — in 1932 most Ukrainian collective households were on served farms — and across regions its coverage rises with the Ukrainian population share, so a region-level comparison cannot separate what the station took from what Ukraine's other burdens took, a collinearity Section 7 returns to when extraction is set against mortality. The dynamic surveys make the separation within region: they report the harvest's division for served and unserved farms in each region, in 1932, the harvest whose division preceded the mortality, and in 1933, the first under the fee in kind. What the station's claim took, and what the reform added to it, is estimated on those cells (Section 5.1).

The mechanism is visible only where farm books survive. The audited farm accounts of Sverdlovsk oblast, 1933 and 1934, show what the fee was assessed on — the farm's own crop while the law priced a share, the district norm when it priced the hectare, with only the charges

collected at the station's own machines, where the grain could not be hidden, still proportional (Section 5.2) — and what a served farm surrendered in kind, the margin the fee's designer had demanded in advance (Section 5.3). The statutes of 1934 close the account: the delivery norm carried relief for a failed crop; the fee schedule carried none (Section 5.4).

5.1 The in-kind fee and total extraction

The conversion of the fee could have left the state's take unchanged. The same winter's delivery law wrote lower norms for the farms a station served (Section 2), so the new claim in grain arrived together with a cut in the old one; if the cut matched the fee, the reform renamed a price. Whether it did is a question about the sum of the two claims, and the sum is observable: the surveys print the division of the harvest for served and unserved farms of the same region on both sides of the change.

Let E_{grt} be the share of the gross harvest taken from group $g \in \{\text{unserved}, \text{served}\}$ of region r in year t — by the obligatory delivery alone, by delivery and fee together, or distributed to members:

$$E_{grt} = \beta_1 \text{Served}_g + \beta_2 (\text{Served}_g \times \mathbf{1}[t=1933]) + \lambda_r + \mu_t + \varepsilon_{grt}, \quad (5.1)$$

on the eighty cells of the survey: twenty regions, two service groups, 1932 and 1933. Region and year effects hold the comparison within region; standard errors cluster by region, and with twenty clusters the interaction is also bootstrapped. β_1 is the served farm's gap in the money era; β_2 is what the change of form added. A renamed price, fully offset by the delivery cut, gives $\beta_2 = 0$ in the total.

Table 9 reports the estimates of equation (5.1). In 1932, under the money fee, a served farm already surrendered 2.18 points of its gross harvest more than an unserved neighbour — the claim ran through the higher delivery schedule — and its members received 2.95 points less. The reform enlarged the gap. In 1933 the served farm's delivery fell by 6.12 points relative to the unserved, the cut the law had written, but the fee in kind rose by 10.6: the total take rose by $\beta_2 = 4.48$ points of the gross harvest (bootstrap $p = 0.003$). Net of the delivery cut, a served farm now surrendered 6.7 points more than its neighbour, and its members received 5.1 less. Weighting regions by their number of farms or by their sown area raises the rise to 5.02 and 7.05; served farms are larger and the survey's cells are not closed populations (Section 3.3), but the differential change in farm size adds nothing to the interaction.

The two claims were administered as one purse. Across the twenty regions the delivery share of served cells fell by 8.19 points between the two years against 2.07 on unserved cells, and where the fee rose more the delivery was cut more — a correlation of -0.59 across the served cells. The cut was real; it returned little more than half of what the fee took. Within the seven Ukrainian oblasts

Table 9: The served–unserved gap before and after the fee in kind.

Dependent variable (pp of gross harvest)	(1) Delivery	(2) Total take (delivery + in-kind fee)	(3) To members
Served (β_1 , the 1932 gap)	2.18** (0.89)	2.18** (0.89)	−2.95*** (0.67)
Served $\times$ 1933 (β_2)	−6.12*** (1.47)	4.48*** (1.48)	−2.17** (1.01)
Net served gap, 1933 ($\beta_1 + \beta_2$)	−3.9	6.7	−5.1
Wild-cluster bootstrap p (β_2)	< 0.001	0.003	0.017
Region and year fixed effects	Yes	Yes	Yes
Region cells (= observations)	80	80	80

Notes. One regression per column on the region $\times$ {unserved, served} $\times$ {1932, 1933} panel of the dynamic surveys, equation (5.1); 20 regions, cluster-robust standard errors by region in parentheses, wild-cluster bootstrap (Rademacher, region clusters, null imposed) for the interaction. In 1933 served farms' obligatory delivery falls (column 1) but the in-kind fee over-replaces it, so the total take rises (column 2) and members receive less (column 3). Weighting regions by their number of collective farms or by sown area moves the column-(2) interaction to 5.02 and 7.05; dropping the four regions whose unserved cell holds fewer than sixty farms, all Ukrainian, moves it to 5.25; within the seven Ukrainian oblasts alone it is 2.09 (s.e. 2.70), not significant. Adding the served/unserved differential change in sown area per farm, or log farm size, leaves the interaction unchanged to two decimals. Coverage: in the 1932 consolidated accounts 78.6 per cent of Ukrainian collective-farm households were on served farms; in the 1933 survey the served share of households runs from 55.9 (Chernihiv) to 95.0 per cent (Odesa), above the served share of farms almost everywhere. Both coverage figures are the sources' own frames — farms that filed reports, farms the survey drew. *Source:* RGAE 1562 op. 7775; Table 3.

alone the interaction is 2.09 with a standard error of 2.70 — the unserved cells there are thin, four of them under sixty farms — and dropping the four thinnest raises the survey-wide estimate to 5.25: the rise is not a Ukrainian artifact, and the Ukrainian cells alone cannot measure it.

A claim that grows while the claim beside it is being cut is drawing on something the other cannot reach. What the fee was written on — and why that base could yield more exactly where the delivery had to give ground — is what the farm accounts show.

5.2 What the fee was assessed on

What a claim is written on decides who bears the harvest's risk, and the farm accounts can settle which base the station's bill actually followed. In 1933, under the share decree, the fee followed the farm's own crop one for one: an elasticity of 1.20 within district (s.e. 0.38; 93 farm-years in 31 districts, the farms that paid in kind), rejecting zero and not rejecting one, while the delivery, put through the same regression on the same audit, sits an order of magnitude lower (Section 4.3). In 1934 the statute's own arithmetic predicts 0.45 — the proportional threshing and combine charges'

share of the mean bill — and the estimate is 0.40 (s.e. 0.28; 124 farm-years), now rejecting one: where verification was free, at the station's own machines, the base stayed on output, and where it was not, it went to the hectare. The base explains the rise of Section 5.1 — the fee could take more exactly where the delivery had to give ground, because it was written on output the state could see, at the one point in the countryside where it could see it — and it dates the system's turn: on the same audited farms the elasticity of the total take in the farm's own yield is 0.135 in 1933 and 0.027 in 1934, so after one harvest under the new fee the state's whole claim had left the farm's own crop. Online Appendix B reports the estimates, the statutory prediction, and their robustness.

5.3 The served–unserved gap in the farm accounts

The demand Chernov's memorandum made was a quantity: a farm the station served was to surrender more grain from each hectare than its unserved neighbour, whatever the work performed (Section 2), and the audit's accounts, which carry every claim the state collected in grain, can read that margin off the books of the farms that paid it. In 1934, across 205 farm-years over 44 districts, a farm served in both years surrendered 3.27 centners a hectare against 2.92 on a never-served farm: +0.35 (s.e. 0.11, clustered by district), which is 2.2 puds — the two puds the memorandum had demanded — while grain distributed to members mirrors the take, 0.43 centners a hectare less on the fully served farms (s.e. 0.18). The farms the station reached only in 1934 separate the fee from the delivery cut: the station billed fewer operations in a first season, the lower delivery norm covered the smaller fee, and their total take stands at the never-served level (−0.01, s.e. 0.09); the surplus was surrendered where the station had worked longest and billed most. Online Appendix B reports the group accounts in full. The margin was measured here on the average served farm in a recovery year. What the claims did where the crop failed — the famine's own case — accounts from a recovery year cannot show. The two statutes can, and they answer it differently.

5.4 The statutory incidence of the two claims

The answers stand in the schedules themselves, and no farm data enter them. By 1934 both claims were written on the hectare, but the two laws built their rates differently: the delivery norm carries no yield term, while the fee's kilograms rise with the district's assessed band. Dividing each printed rate by the crop it claims shows what each law takes at each level of the harvest.

Table 10 reports the arithmetic at the midpoint of each band of the 1934 fee schedule. The delivery norm's effective share falls from 90 per cent of a failing crop to 13 per cent of a good one at the Ural rate for served farms, and from 120 to 17 at the Ukrainian; the 1933 fee is 20 per cent at every yield by construction; the 1934 fee rises from 14.5 to 21.6 per cent across the bands. These are properties of the printed rates, not estimates: a fixed quantity takes a larger share of a

smaller crop, and the 1934 graduation reverses that slope only between bands. Read together, the two schedules divide the yield — the delivery's share is largest where the harvest is worst, the fee's where it is best.

At a failed crop the two laws differ in what they provide. The 1933 fee, a share, shrinks with the crop by construction. The 1934 law has no such property and no substitute for it: the decree contains no provision for crop failure, and the threshing charge was owed even where the tractors had not worked (Section 2). Its band was assigned to each farm by a district commission — the station's director sitting on it — within twelve days of the start of harvest, before threshing results existed (Section 2). The delivery, by contrast, was adjusted by administration where harvests collapsed: the quota cuts of Section 4.1. No source in this study records the bands the commissions assigned, so whether failing farms were placed above their realised crops cannot be tested; the printed rates make it possible — at two centners under a band of seven to nine, they take 53 per cent of the crop.

The asymmetry is the documentary finding: of the two claims written on the 1934 harvest, one carried a provision for crop failure and the other did not. The pattern of bases points the same way — proportional charges survived where the station's own machines handled the grain, assessed quantities replaced them elsewhere — the arrangement a state would choose if it matched each claim's form to what it could observe. Over-charging through the band assignment remains a conjecture: the printed rates permit it, and no surviving source measures it. Section 4.8 compared a proportional rule with a fixed one as a counterfactual; in the statutes of 1934 the two stood side by side in law.

6 Grain Retained and Famine Mortality

What a collective farm kept after the state's claims was the food its members lived on until the next harvest. The famine mortality of 1933 therefore admits two explanations on the production side: harvests too small to feed the farm population, or claims that removed too much of an adequate harvest. Both quantities are observed. For the same collective farms in each region, the surveys report the gross grain harvest and the grain distributed to members, and the mortality data described in Section 3 give each region's death rate in the year that followed.

Section 6.1 compares the share of farms below a subsistence line on their own harvest with the share below the same line on the grain they retained, and finds that only the second share predicts the rise in the death rate. Section 6.2 measures the depth of the shortfall below the line and shows that it explains the cross-region gradient of mortality better than the extraction rate itself, which is informative only through the residual it leaves. Section 6.3 examines Kyiv and Sverdlovsk, two regions with the same extraction share but very different residuals and death rates, and treats

Table 10: The share of the harvest taken at each yield, by statute.

assessed band (c/ha)	midpoint (c/ha)	delivery, Urals (No. 78)	delivery, Ukraine (No. 78)	fee 1933 (No. 346)	fee 1934 (No. 390)
<3	2	90%	120%	20%	14.5%
3-<5	4	45%	60%	20%	15.5%
5-<7	6	30%	40%	20%	18.0%
7-<9	8	22%	30%	20%	19.2%
9-<11	10	18%	24%	20%	20.0%
11-<13	12	15%	20%	20%	20.5%
13+	14	13%	17%	20%	21.6%

Notes. Statute on statute; no farm data. Each cell divides a law's printed rate by the band-midpoint yield (open bands evaluated at 2 and 14). Delivery: the fixed rye norm of decree No. 78 for MTS-served farms (Urals 1.8, Ukraine 2.4 centners per hectare) — the schedule contains no yield term, so its share is arithmetic. Fee 1933: the bundled 20 per cent of decree No. 346, threshing included — a share, so flat by construction. Fee 1934: decree No. 390's full-service complex in kg/ha at each band plus the compulsory 8 per cent threshing toll; the band is assigned to each kolkhoz by a district commission within twelve days of the start of harvest (item 12), so a farm whose barn crop lands under its band owes the band's quantity — at 2 centners under a 7–9 band the schedule takes 53 per cent of the crop. No. 390 contains no crop-failure clause. *Source:* SZ SSSR 1933 No. 9 st. 47, No. 4 st. 25; 1934 No. 11 st. 68.

Sverdlovsk — a low residual with little excess mortality — as the principal case against the argument. Section 6.4 carries the residual below the region: the printed distributions bound every district-level arrangement of the residual consistent with them, the mortality gradient survives each one, and the surviving slope converts the division into a count of deaths. The residual accounts for the incidence of subsistence failure, but not for the higher mortality of Ukrainian districts, which is measured at the district level, where the residual is not observed; Section 7 takes up that gradient.

6.1 Harvest failure and the division of the harvest

Whether a region's farms were short of food because their harvests were small or because the claims on those harvests were large cannot be decided from mortality data alone: the two causes produce the same outcome. The surveys, however, record each farm's grain at two points in the same accounting chain — the gross harvest, and the amount distributed to members after the state's claims and the farm's seed and fodder — so the share of farms below subsistence can be computed at both points and the difference attributed to the division.

Let H_r^{grown} denote the share of region r 's collective farms whose gross grain harvest of 1932 was below 1,600 kcal per soul per day — the working ration used in Section 4.7, converted by equation (4.5) — and H_r^{left} the share below the same line on the grain actually distributed to members. The difference

$$W_r = H_r^{\text{left}} - H_r^{\text{grown}}, \quad (6.1)$$

the wedge, measures the share of farms brought below subsistence by the division rather than by the harvest.² Table 11 reports both shares, their difference, and the change in the rural death rate for the 19 surveyed regions for which a 1928 death rate exists. Harvest failure was rare: in the median region, 0.0 per cent of farms produced less than the working ration, while 71.4 per cent were below the line on the grain they received; the median wedge is 60.1 percentage points. The attribution does not depend on the choice of line: below the survival threshold used in Section 4.8 the count on the farms' own harvests is close to zero everywhere, so nearly every farm under that threshold was placed there by the division.

In a regression of the 1928–1933 change in the rural death rate per 1,000 on each measure separately — 19 regions, standardised regressors, heteroscedasticity-robust standard errors — harvest failure is not a significant predictor: +11.50 deaths per 1,000 per standard deviation, with $t = 1.08$ and $R^2 = 0.227$. The wedge of equation (6.1) is a significant one: +13.10, with $t = 4.73$ and $R^2 = 0.295$. When both are included, the coefficient on the wedge is essentially unchanged at +13.73 ($t = 4.16$), harvest failure remains insignificant (+12.21, $t = 1.12$), and the joint R^2 is 0.551. The two measures are uncorrelated across regions ($r = -0.05$), so the wedge is not a proxy for the severity of the harvest.

The two relationships are plotted in Online Appendix B: the change in the death rate is unrelated to the share of farms whose harvests failed, and increases steeply with the share left below the line after distribution. Because the comparison uses the same farms' grain at two points in the same accounting chain, within region, the result attributes the subsistence failure of 1932 to the division of the harvest rather than to its size.

6.2 The shortfall and the rise in mortality

A headcount weights every farm below the line equally: a farm short of the ration by a tenth counts as much as a farm left with almost nothing. If mortality responds to the severity of deprivation rather than to its incidence alone, the appropriate measure is the depth of the shortfall — the average distance below the subsistence line across a region's farms.

²The constructed residual can be compared with directly measured consumption. The household budget survey of 1933–34 records food consumption from all sources in seven of the surveyed oblasts, listed in Table 3. In the first half of calendar 1933, which was lived on the residual of the 1932 harvest, measured intake was lowest in the two oblasts where the constructed residual is lowest — Kyiv, 1,212, and Odesa, 1,147 kcal per person per day, against 1,782–2,356 in the other five — and Kyiv's measured grain consumption of 705 kcal per day is within 10 per cent of its constructed kolkhoz grain residual of 635. The cross-oblast orderings for the famine year are positively correlated (Pearson +0.65 to +0.69; Spearman +0.39; $n = 7$); for calendar 1934 the correlation disappears, which is consistent with the residual constraining consumption only near subsistence — above it, households sold grain and consumed from the household plot and the market. The collective distribution supplied a median of 80 per cent of consumed calories in calendar 1933, and 43 per cent in Kyiv.

Table 11: Farms below subsistence on the harvest and on the distribution, 1932.

	% of farms below the line, 1932		wedge	rise in death rate per 1,000, 1928→1933
	on own harvest	on what was left		
Kharkiv [†]	0.0	98.9	98.9	+55.7
Donetsk [†]	0.0	90.5	90.5	+21.8
Azov–Black Sea	0.0	88.7	88.7	+42.1
Odesa [†]	0.0	88.5	88.5	+36.4
Dnipropetrovsk [†]	0.0	83.7	83.7	+56.3
Vinnytsia [†]	5.6	88.8	83.2	+47.3
Sverdlovsk	0.0	68.3	68.3	+7.1
Leningrad	3.4	71.4	68.0	+1.9
CBE	5.7	71.9	66.2	+14.1
Stalingrad	0.0	60.1	60.1	+42.1
Chernihiv [†]	33.3	87.7	54.4	+32.2
Kyiv [†]	43.1	94.3	51.2	+82.6
Moscow	2.5	46.3	43.8	+1.6
Ivanovo	1.9	44.6	42.7	+4.0
Western	5.7	45.9	40.2	-1.6
Tatar ASSR	0.0	18.9	18.9	+9.1
W. Siberia	0.0	15.9	15.9	+3.7
Mid-Volga	0.0	14.3	14.3	+11.4
Gorky	0.4	12.3	11.9	+5.2
Median	0.0	71.4	60.1	

Notes. Region level, the dynamic surveys, 1932 harvest year, all 19 surveyed regions with a 1928 death rate (BSSR lacks one and is absent). The subsistence line is 1,600 kcal per soul per day in grain. Both headcount columns count the same collective farms of a region against that line: first as if the members ate the farm’s whole gross grain harvest, then on the grain actually distributed to them after the state’s claims and the farm’s reserves; the wedge is the difference. The death-rate column is the region’s rural crude death rate in 1933 less 1928, per 1,000, aggregated from district rates as described in Section 3. [†] Ukrainian oblast. *Source:* Tables 3 and 4.

For region r the shortfall is

$$D_r = \frac{1}{N_r} \sum_i \max(0, s - x_i), \quad (6.2)$$

where x_i is the grain distributed per soul on farm i and the line s is 1.5 centners per soul for the year, equivalent to roughly 1,400 kcal per day and slightly below the working ration of Section 6.1. The surveys do not report x_i farm by farm; they report two groupings of the same farms, by yield and by sown area per worker, and D_r is therefore computed from the two marginal distributions, with the unobserved coupling between them bracketed by its Fréchet bounds. The estimates reported first use the central case of independence; the bounds accompany the robustness checks. The surveys cover 20 regions, 19 have a 1928 death rate — the Belorussian republic has none — and

the construction exists for 18 of the 19. In a regression of the 1928–1933 change in the rural death rate on the shortfall, with the standardisation and standard errors of Section 6.1, the coefficient is +18.89 deaths per 1,000 per standard deviation, with $t = 4.29$ and $R^2 = 0.580$.

The alternative measures are weaker. The headcount below the same line gives +16.17 ($t = 2.96$, $R^2 = 0.425$). Mean grain per soul, entered linearly, gives -16.14 on its own ($R^2 = 0.423$) and reverses sign to an insignificant +9.93 ($t = 0.76$) when the shortfall of equation (6.2) is added: once the depth of deprivation is accounted for, the average payout contains no further information. The delivery share of the gross harvest — the extraction rate itself — gives +12.99 on its own ($R^2 = 0.290$, $n = 19$) and +6.82 ($t = 1.17$) against the shortfall, so the rate predicts mortality only through the shortfall it produced. The caloric headcount of Table 11, constructed from printed tables different from those behind the shortfall, gives +16.86 ($t = 4.32$, $R^2 = 0.488$) on the 19 regions.³

One distinction the regional cross-section cannot make must be stated. Across the 18 regions the mean payout and the shortfall are correlated at -0.928 (Spearman -0.979), a variance-inflation factor of 7.2, and a comparison restricted to regions with nearly identical means — paired to within 8 per cent — yields a shortfall coefficient of only +0.155. The convex measure fits better than the linear one, but on 18 regions the data cannot establish that the depth of deprivation mattered beyond the level of the payout. The identified result remains that of Section 6.1: the shortage was created by the division, not by the harvest.

The estimate is stable under the construction's assumptions. At the two Fréchet bounds on the coupling the coefficient is +20.71 and +18.16 ($R^2 = 0.697$ and 0.536), against +18.89 at independence. At subsistence lines of 1.25 and 1.75 centners per soul it is +19.28 and +18.72. On the alternative construction of the shortfall, from the sown-per-worker grouping alone, it is +19.84 ($t = 5.21$, $R^2 = 0.639$). Omitting one region at a time moves it between +14.18 and +20.43; Kyiv is the most influential observation. Figure 3 plots the headcount and the shortfall against the change in the death rate.

The pooled gradient differs inside the two halves of the panel. Within the twelve RSFSR regions the change in the death rate is most closely associated with the extraction rate (correlation +0.809; +12.01 per standard deviation, $t = 2.92$). Within the six Ukrainian oblasts for which the shortfall exists it is most closely associated with the shortfall (correlation +0.825; +14.89, $t = 1.98$), and the delivery share is negatively correlated with mortality across the seven Ukrainian oblasts (-0.252). Samples of six and twelve support description rather than separate estimates, but the Ukrainian pattern has a concrete source: the oblast with the highest mortality surrendered

³Measured in calories, the served–unserved gap of Section 5 is also larger than in grain: counting the distributed potato and sunflower with the grain, the members of a served farm received on average 110 kcal per soul per day less than those of an unserved farm in 1932, and 127 less in 1933, against gaps of 80 and 92 on grain alone. These figures cover the collective distribution only and are lower bounds on food from all sources.

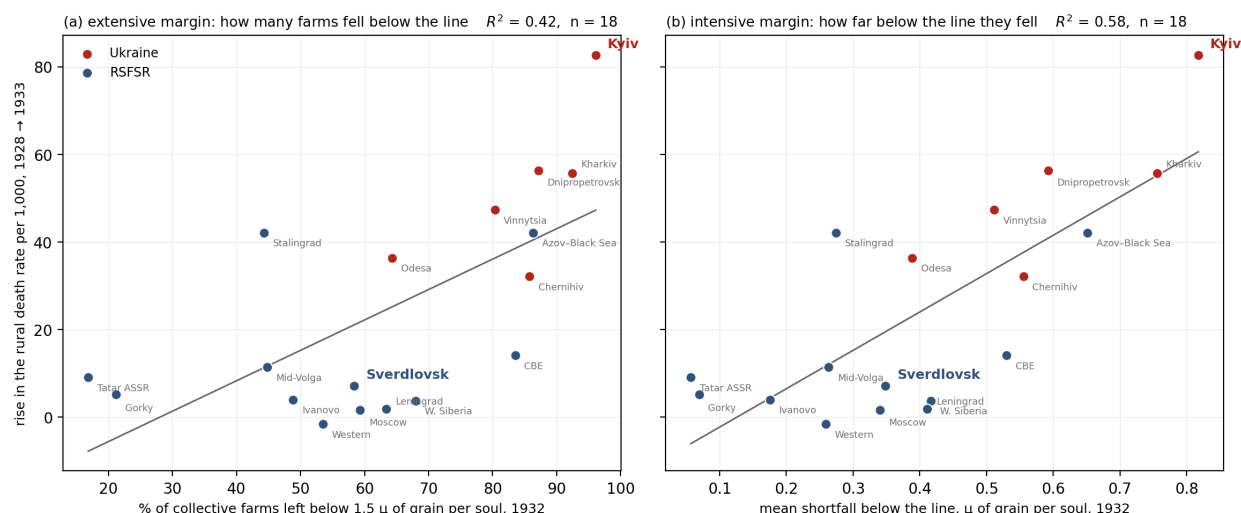


Figure 3: Mortality and the shortfall below subsistence.

Notes. Region level, $n = 18$ surveyed regions with the joint construction. The vertical axis in both panels is the change in the rural death rate per 1,000 between 1928 and 1933. The horizontal axis is, in the left panel, the share of collective farms left below 1.5 centners of grain per soul in 1932, and in the right panel, the mean shortfall below that line in centners per soul. The unstandardised slopes are +0.694 deaths per 1,000 per percentage point (left) and +8.76 per 0.1 centner per soul (right). Lines are OLS fits. *Source:* the dynamic surveys and the district death rates, Tables 3 and 4.

an ordinary share of a very small harvest, the case examined in Section 6.3.

6.3 Kyiv and Sverdlovsk under the same claim

A pair of regions makes the estimates concrete. The state's claim, expressed as a share of the harvest, could be nearly identical in two regions and still leave their populations in different conditions, because what a household received depended on the harvest from which the share was taken. The same pair supplies the section's principal counterexample: a region whose farms the accounts place far below the line, with little excess mortality.

Table 12 compares Kyiv and Sverdlovsk oblasts in 1932 — the Ukrainian oblast with the highest rise in the death rate, and the oblast of the farm audit described in Section 3. The two surrendered almost the same share of their gross grain harvests, 29.0 and 28.9 per cent, and distributed similar shares to their members. The harvests behind those shares differed: Kyiv's farms held 1.06 sown grain hectares per worker against Sverdlovsk's 2.53, so the same proportional claim left Sverdlovsk's members 1.58 centners of grain per soul — more than twice the Kyiv figure reported in Section 4.1 — or 1,468 kcal per soul per day against Kyiv's 655. The subsistence measures follow: 96.1 per cent of Kyiv's farms below the 1.5-centner line, with a mean shortfall of 0.817 centners per soul, against 58.4 per cent and 0.348 in Sverdlovsk. Kyiv's rural death rate rose from 16.2 per 1,000 in 1928 to 98.9 in 1933; Sverdlovsk's from 26.8 to 33.9.

Table 12: Kyiv and Sverdlovsk oblasts, 1932.

	Kyiv	Sverdlovsk
Barn yield, centners per hectare	4.49	5.30
Grain sown area per worker, hectares	1.06	2.53
Obligatory delivery, per cent of the gross harvest	29.0	28.9
Distributed to members, per cent of the gross harvest	27.0	23.9
Distributed grain, kcal per soul per day	655	1,468
Farms below 1.5 centners per soul, per cent	96.1	58.4
Mean shortfall below the line, centners per soul	0.817	0.348
Rural death rate per 1,000, 1928	16.2	26.8
Rural death rate per 1,000, 1933	98.9	33.9

Notes. Region level, the dynamic surveys, 1932 harvest year; death rates from the district mortality data of Section 3. The kcal row converts the grain distributed to members by equation (4.5); the two subsistence rows use the 1.5-centner line of equation (6.2). Kyiv's grain per worker and per soul are reported in Section 4.1; the caloric headcounts and the change in the death rate are reported in Table 11.

Source: Tables 3 and 4.

The configuration was visible to the government at the time. Kosior's report to the Central Committee of 15 March 1933 asked why famine was concentrated in Kyiv oblast, where, in its words, almost no grain had been taken by procurement in its principal districts (Danilov et al., 2001, doc. 274). The report's explanation — local mismanagement of the distribution — served its author, who answered to Moscow for the republic's procurement; the delivery data of Section 4, not his account, measure what was taken. The report's allocation figures are the informative element. Of the 200 thousand puds of food aid it allocated from the republican reserve, Kyiv oblast received 140 thousand, with the instruction that at least half go to the sugar-beet districts to secure the working of the beet: relief was tied to the spring field work, not to the shortfall.

Sverdlovsk is the case that does not fit. The accounts place 68.3 per cent of its farms below 1,600 kcal per soul per day on the collective distribution (Table 11), and its death rate rose by 7.1 per 1,000 — far below the fitted line of Figure 3, and below that of the corresponding plot in Online Appendix B. Two explanations are open, and no source in this study measures either for the oblast. First, every figure counts the collective distribution only: food from the household plot, earnings outside the farm, and purchases are absent from the accounts, and where the budget survey of Section 3 does measure these margins it finds them substantial. Second, the audited farms of the same oblast show the response to a low residual in their population rather than in regional deaths: a farm's 1934 population rises with the food it distributed, an elasticity of +0.135 on the 272 audited farms — an adjustment the farm accounts cannot divide between deaths and departures, and departures do not enter a regional death register. The counterexample bounds the claim: a low collective residual was compatible with modest excess mortality where food outside the collective

distribution existed. The regressions of Section 6.2 establish the gradient across regions, not a prediction for each region.

6.4 The residual below the region level

The residual is measured at the level of the region, and deaths are registered by district. A gradient estimated across regions is therefore open to an objection from aggregation: regional averages are compatible with many arrangements of the districts beneath them, and some arrangement might exist under which the gradient would not survive. The distributions the surveys print for each region make the objection answerable, because they bound how far any district arrangement can depart from its regional average.

Each region's survey volume prints, beside the mean, the distribution of its collective farms over grain per labour-day, and the two printed means on the same row convert its bin edges to grain per soul. A lognormal distribution with the printed mean G_o and a spread σ_o fitted to the printed bins describes region o 's farms, and a district arrangement consistent with it assigns district r the mean

$$\ln g_r = \ln G_o - \frac{1}{2} \theta \sigma_o^2 + \sqrt{\theta} \sigma_o \left(\rho z_r + \sqrt{1 - \rho^2} \varepsilon_r \right), \quad (6.3)$$

rescaled so that the population-weighted district means reproduce G_o exactly. Here z_r is the district's standardised log quota base — procurement per soul in the Ukrainian districts, sown area per soul in the two RSFSR regions, the base of the per-hectare norms of Section 4 — and ε_r is noise. The two parameters no source measures are swept rather than assumed: θ , the share of the variance lying between districts, over 0.25–0.60, and ρ , the correlation of a district's residual with its quota base, over -0.75 to $+0.75$, so that the grid contains both the reading in which the quota fell on the well-provided districts and the reading in which it stripped them. Every admissible arrangement yields, in closed form, each district's share of collective farms below one centner of grain per soul for the year, H_r — a stricter line than that of equation (6.2).

The gradient survives every arrangement. In a regression of the district's 1928–1933 change in the crude death rate per 1,000 on H_r and urbanisation, population-weighted, without region intercepts, on the 493 districts of the three regions with a district quota base — 356 Ukrainian, 75 Ural, 62 Lower Volga — the coefficient on H_r lies between $+11.5$ and $+16.0$ deaths per 1,000 per standard deviation over the whole grid of admissible arrangements ($p = 0.006$ at the central cell, standard errors clustered on the nine regions); the district mean g_r in place of the headcount gives -16.3 to -11.6 . The Lower Volga enters as the test case — its rise of $+40.2$ per 1,000 is the largest outside Ukraine — and moves the central coefficient only from $+15.1$ to $+13.3$. With region intercepts the coefficient lies between -3.5 and $+4.0$ and is never significant, and within the Ukrainian oblasts no admissible arrangement explains more than one per cent of the within-region

variance of district mortality. The districts add no measurable food information of their own, so nothing measurable is lost by working at the region level — the level at which the delivery rates of Section 4 themselves varied.

The gradient belongs to the 1932 campaign, not to standing regional characteristics. Recalibrated to the printed distributions of the 1933 harvest — the harvest that arrived after the mortality peak of 1933 — the association disappears: across the seven Ukrainian oblasts the rank correlation of the regional headcount with district mortality falls from +0.57 to +0.07, and its coefficient in the corresponding district-level regression from +46.5 per standard deviation ($p = 0.02$) to an insignificant +8.8. Both distortions of the 1933 figures run the other way: the per-soul denominators of 1933 count survivors, and persistence alone would have carried the 1932 ordering forward.

A slope in natural units converts the gradient into deaths. Re-estimated with the headcount in percentage points, the regression assigns each additional point of collective farms below the line between 0.41 and 0.45 deaths per 1,000 across the grid. Holding every district at a reference residual and summing the predicted change over district populations gives the excess deaths attributable to the division,

$$\Delta = \hat{\beta} \sum_r N_r (H_r - H^{\text{ref}}) / 1,000, \quad (6.4)$$

and Table 13 reports Δ , the implied count of deaths, against two references. Held at the Sverdlovsk residual — the least deprived region inside the frame — the division accounts for 302–479 thousand of the frame’s 1,314 thousand excess deaths, or 23–36 per cent; held at the Gorky residual, the least deprived among all twenty surveyed regions, for 581–813 thousand, or 44–62 per cent. The brackets are narrow because the conversion runs on the printed between-region differences, which every admissible arrangement preserves; the unidentified within-region allocation nearly cancels in the population-weighted sum. The counterfactual alters the division of the harvests actually gathered, not the harvests: the farms it moves above the line are those Section 6.1 shows were placed below it by the division. Four restrictions accompany the estimate. The slope is a cross-district association, applied linearly. The excess deaths are the one-year rise of the death rate, not the famine’s full toll. The frame is three regions of the twenty surveyed, and the Ukrainian districts contribute 1,166 of the 1,314 thousand. And the headcount counts the collective distribution only, so deprivation — and with it the attributed share — is understated wherever the household plot fed people.

7 The Within-Region Mortality Residual

The rules of Sections 2 and 4 are written on sown area, past deliveries, and the classifications of region and service status; none refers to nationality, and the regional gradient of extraction is

Table 13: Excess deaths attributable to the division.

counterfactual residual	deaths, thousands	share of the frame's excess deaths
every district at the Sverdlovsk residual	302–479	23–36 per cent
every district at the Gorky residual	581–813	44–62 per cent

Notes. District level, $n = 493$ districts in the three regions with a district quota base (356 Ukrainian, 75 Ural, 62 Lower Volga); the frame's total excess deaths — the 1928→1933 rise in the rural crude death rate times rural population, summed over districts — are 1,314 thousand. Each bracket is the minimum 2.5th to the maximum 97.5th percentile of the implied counterfactual over the full grid of admissible allocations (both unknown parameters swept, 200 draws per cell); the slope behind the conversion is 0.41–0.45 deaths per 1,000 per percentage point of collective farms below one centner of grain per soul. The Sverdlovsk row holds every district at the least-deprived residual inside the frame; the Gorky row at the least-deprived residual among all twenty surveyed regions. *Source:* Tables 3 and 4, and the district data of Section 3.

carried by the producing-status schedule (Section 4.3). The division those rules produced accounts for the between-region layer of famine mortality (Section 6). What it leaves unaccounted for is a gradient inside the regions: across districts of the same oblast, mortality rises with the Ukrainian population share of the 1926 census. The gradient sits exactly where the sources thin out. Grain retained is printed one level above, by region, so the food variable of Section 6 and the population share never meet at the district level (Section 3.6).

The decomposition locates each variable's contribution (Table 14). Of the variance of the 1928-to-1933 rise in district death rates across the 355 Ukrainian districts of the mortality panel, 35 per cent lies between the seven oblasts and 65 per cent within them; the regional share of farms left below subsistence accounts for about two thirds of the between part, 23 per cent of the total — the layer at which the claims of Sections 4 and 4.3 were set — while the Ukrainian share adds 4.2 per cent of the total, all of it within regions, where the food allocation is unobserved and where relief conditionality, the restrictions on movement, and local enforcement are unmeasured. Section 7.1 bounds what the unobserved district food allocation can contribute to the within-region gradient: under every allocation the printed distributions admit, including allocations targeted on the population share itself, the gradient stands. Section 7.2 measures what the residual can be instead — its magnitude expressed in deaths, the recorded requisition and relief practices that do not carry it, its structure across the other nationalities, and the documented instruments outside the grain apparatus that remain.

7.1 Bounds on confounding by the unobserved food allocation

A mortality gradient estimated on district population shares, with district food unobserved, is open to a confounding objection: if the shortfall was allocated within each region toward the districts where the titular population was concentrated, the share would carry a food effect and measure

Table 14: The two layers of district mortality.

<i>Panel A. Where the variance sits: the rise in the district death rate, 1928–1933</i>	
	share of total variance
Between the seven oblasts	35 per cent
of which: carried by the regional share of farms below subsistence	23 per cent
Within oblasts	65 per cent
of which: added by the Ukrainian population share (ΔR^2)	4.2 per cent
<i>Panel B. The identified sets, over every admissible district food allocation</i>	
Object	Set
Farms below subsistence → mortality, between regions (no region intercepts; 493 districts, population-weighted)	[+11.5, +16.0]; excludes zero
The same, within regions (region intercepts)	[−3.5, +4.0], n.s.; ≤ 1 per cent of within-region variance
Ukrainian population share, within oblasts (355 districts, oblast fixed effects, population-weighted)	[+24.5, +26.9]
Recalibrated to the 1933 harvest	food +46.5** → +8.8 n.s.; the population share unchanged

Notes. Panel A decomposes the variance of the 1928-to-1933 rise in district death rates across the 355 Ukrainian districts of the mortality panel; the regional below-subsistence share accounts for about two thirds of the between-oblast part. Panel B reports, for each coefficient, the range it takes over the whole grid of district food allocations consistent with the printed regional distributions, in deaths per 1,000 per standard deviation; the first two rows and the food half of the last restate the estimates of Section 6.4, while the third row, and the invariance of the population share in the last, are established in this section. The last row recalibrates the food headcount to the printed distributions of the following harvest across the seven Ukrainian oblasts. *Source:* Tables 3 and 4, and the district data of Section 3.

deprivation rather than anything attached to nationality. No district food measure exists to hold fixed, so the objection cannot be answered by a control variable. It can be answered by bounds. The surveys print each region's mean and its distribution of collective farms over grain per labour-day, and those printed moments cap both how much food dispersion any district allocation can contain and, through the mortality regressions, how much of the gradient such an allocation can absorb — the identification problem of a regression whose omitted variable is known only through group-level distributions (Cross and Manski, 2002).

The observable part of the allocation is not tilted toward Ukrainian districts. The quota base of Section 4 — procurement per rural soul, standardized within oblast — has a correlation of −0.045 with the Ukrainian share across the 357 districts of the seven oblasts (−0.043 population-weighted), and the sign is not even stable across oblasts, running from +0.32 in Kyiv to −0.35 in Odesa, the two ends of the range. Whatever the within-region allocation followed, the instrument the state could observe did not follow the population share.

The printed dispersion then bounds what an unobservable tilt could do. An omitted district food variable biases the coefficient on the share by at most the mortality response per unit of food, times the within-region standard deviation of the food variable, times their correlation. The response per unit is the slope behind Table 13; the within-region standard deviation of the district headcount below subsistence is capped by the printed distributions at 27.9 percentage points (population-weighted, at the largest admissible between-district variance share). An allocation perfectly correlated with the Ukrainian share therefore contributes at most 11.4–12.5 deaths per 1,000 per standard deviation of the share, against an estimated gradient of +24.5 to +26.9 (the 355 districts, oblast fixed effects, population-weighted; the range spans every admissible allocation). Overturning the gradient would require a correlation near one and, at the same time, a within-region mortality response at least twice the cross-district slope. Neither is observed: the quota base above and the requisition testimonies of Section 7.2 put every measurable correlate of the allocation near zero.

The bound also holds inside the regressions. Extending the allocation class of Section 6.4 with a targeting axis — the district draw correlated, up to ± 0.95 , with the within-oblast Ukrainian share — and re-estimating the mortality regression on each draw leaves the coefficient on the share between +6.4 and +35.7 across the whole grid on the district loss rates, and between +1.7 and +13.6 on the census-based change in death rates; it crosses zero nowhere. The same result can be stated in sensitivity form (Cinelli and Hazlett, 2020): to drive the coefficient to zero an omitted variable would need to explain at least 24 per cent of the residual variance of both the share and mortality; allocations targeted on the share reach 71 per cent on the first axis but never exceed 5 per cent on the second — an allocation can be made to track the share, but no admissible allocation predicts the mortality the regression leaves unexplained — and at the most adverse attainable pair the adjusted coefficient is +5.3. Online Appendix C plots the sensitivity contours and the attainable pairs.

The two variables also differ in time. Recalibrated to the printed distributions of the following harvest, the food gradient of Section 6.4 disappears, exactly as the timing of the deaths requires; the coefficient on the Ukrainian share does not move. A share that stood in for the food allocation would fall with it.

7.2 The size of the residual, and the channels that remain

The gradient that survives the bounds measures channels the sources do not record. Its reading is disciplined by three measurements: its magnitude expressed in deaths, the recorded practices that fail to carry it, and its structure across the other nationalities.

The magnitude first. Moving every district from the ninetieth to the tenth percentile of the

Ukrainian share — from 0.99 to 0.67 — at the estimated within-oblast gradient lowers the predicted death rate by 53 per 1,000; summed over district populations, the difference is 871 thousand of the 3,530 thousand recorded rural losses of the 355 districts, 24.7 per cent. The number is an upper bound on what unmeasured within-region channels correlated with the share could account for, not an estimate of any named mechanism: it prices the whole residual gradient, whatever produced it.

The recorded practices do not carry it. Nearly two thousand geocoded survivor testimonies, assigned to districts of the 1933 frame, yield district-level shares of testimony reporting grain requisitions, the seizure of food and belongings, livestock seizures, and the receipt of state assistance by its conditionality category; 243 districts holding 77 per cent of the rural population have at least three testimonies. The measures carry real famine signal — a district's loss rate predicts the share of its testimony reporting cannibalism at +0.22 standard deviations per 100 deaths per 1,000 — yet none of them is related to the Ukrainian share within oblasts, and adding all five to the mortality regression moves the coefficient on the share from +27.3 to +29.1 on the covered districts. The requisition channel itself is real: the grain-requisition share predicts district mortality at +19.9 deaths per 1,000 per standard deviation conditional on the share and the quota base. Confiscation on the ground was lethal where it was intense; its intensity was not patterned on the population share. The testimonies are survivor-sampled and recorded decades later, so these are measures of remembered intensity, never administrative counts, and the mortality association may partly reflect the severity of the famine recalled rather than its cause.

The one administrative measure recorded at the district level behaves the same way. Within oblasts, the collectivization rate of late 1932 predicts mortality with opposite signs across the republic — +44.5 deaths per 1,000 per standard deviation in Kyiv and +20.0 in Vinnytsia, against −28.8 in Odesa, −30.7 in Dnipropetrovsk and −35.5 in Donetsk, all robust to spatial correlation — and the sign follows capacity: collectivization predicts death where the per-head quota base was small and the opposite where it was large (+12.0, +5.1 and −10.2 across its thirds). A measured instrument whose association reverses by oblast cannot carry a republic-wide gradient in one nationality.

Nor is the residual a single gradient in one nationality. Re-measured from the district-level 1926 census — 625 districts of the 1927 frame, carried onto the mortality frame by areal interpolation, agreeing with the coarser measure at a correlation of 0.954 — the Ukrainian share gives the same coefficient, so the gradient is not an artifact of measurement. The other shares add structure: conditional on the Ukrainian share, districts with a larger Polish rural share died less by 16.9 per 1,000 per standard deviation, and with a larger Russian share by 13.6; beside them the Ukrainian coefficient falls to +16.2 with a *p*-value of 0.14 on seven oblast clusters. The protective minority differentials concentrate in the right-bank border belt, where consular ties, foreign remittances and

the border regime are candidate readings that no source in this study measures. Whatever the within-region mechanism was, it differentiated by nationality in more than one direction, and none of it ran through an instrument recorded here.

Where an allocation and a repression measure are both observed at fine level, the confounding the bounds guard against does not appear. In the Sverdlovsk farm panel of Section 3.4, a village-council measure of dekulakization victims per 1,000 enters the farm-level relation between food left and survival without moving it, across 28 specifications, and carries no association of its own. The one region where the fine-level test can be run supplies no example of a repression channel masquerading as a food gradient, or the reverse.

The channels that remain are documented, and they are not the quota. The Politburo tied the procurement failure in Ukraine and the North Caucasus to nationality policy in December 1932, restricted peasant movement out of both regions by name in January 1933 (Davies and Wheatcroft, 2004, pp. 187–196), and conditioned the relief of spring 1933 on collective-farm membership and sowing performance (Kosior, 2001). None of these instruments is a function of the delivery schedule, the assessed base, or the station fee, and none is measured at the district level in any source used here. The service-status test of Section 5, identified where the Ukrainian unserved cells are thinnest (Section 3.3), brackets a served-by-Ukrainian differential in the fee between -9.2 and $+6.1$ points of the harvest — a null without power, reported as such. Markevich et al. (2025) attribute approximately half of the Ukrainian mortality effect to bias in the centrally planned procurement policy; at the district level their food-side control is agro-climatic suitability, and food after procurement enters their design only at the province level. The bounds above are the missing step for both studies: within regions, the procurement apparatus and every allocation of food consistent with its records leave the gradient intact, so the bias their estimate attributes to procurement must operate through the instruments outside it. The extraction system explains the layer of the famine at which it was written; beneath that layer it leaves a bounded residual whose instruments lie outside the grain apparatus.

8 Conclusion

The grain claim of the early 1930s was written on what the state could verify. Its base was the sown hectare, the district's own past deliveries, and a yield assessed on the standing crop before the harvest; its size was carried forward from campaign to campaign; and from 1933 a second claimant, the machine-tractor station, held a claim on the same harvest, written first as a share of an assessed crop and then as a quantity per hectare — with a provision for crop failure in the delivery statute and none in the fee's.

A claim of that form could not respond when the harvest failed, and the accounts the state kept

for itself record what followed. The harvest moved and the claim did not, so the movement fell on what farms retained; the farms below subsistence were put there by the division of the harvest, not by their fields; and the served farm surrendered more than its unserved neighbour, the margin the fee's designer had demanded in advance. The food left where policy set it predicts the mortality rise of 1933 between regions under every allocation of the residual the printed distributions admit. Beneath the region — the level at which the lists, the rates and the carry-forward varied — the allocation is unidentifiable, and nothing measurable is lost by estimating where the policy was set. What remains within regions is a bounded residual, invariant to every admissible food allocation, whose candidate instruments — the conditioning of relief, the restriction of movement, local enforcement — the sources do not measure and the quota apparatus does not carry.

The state taxed the base it could observe and accepted the loss; with a subsistence floor beneath the producer, that loss fell on survival. The schedules make no reference to nationality, and the regional gradient of extraction disappears once their terms are accounted for.

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